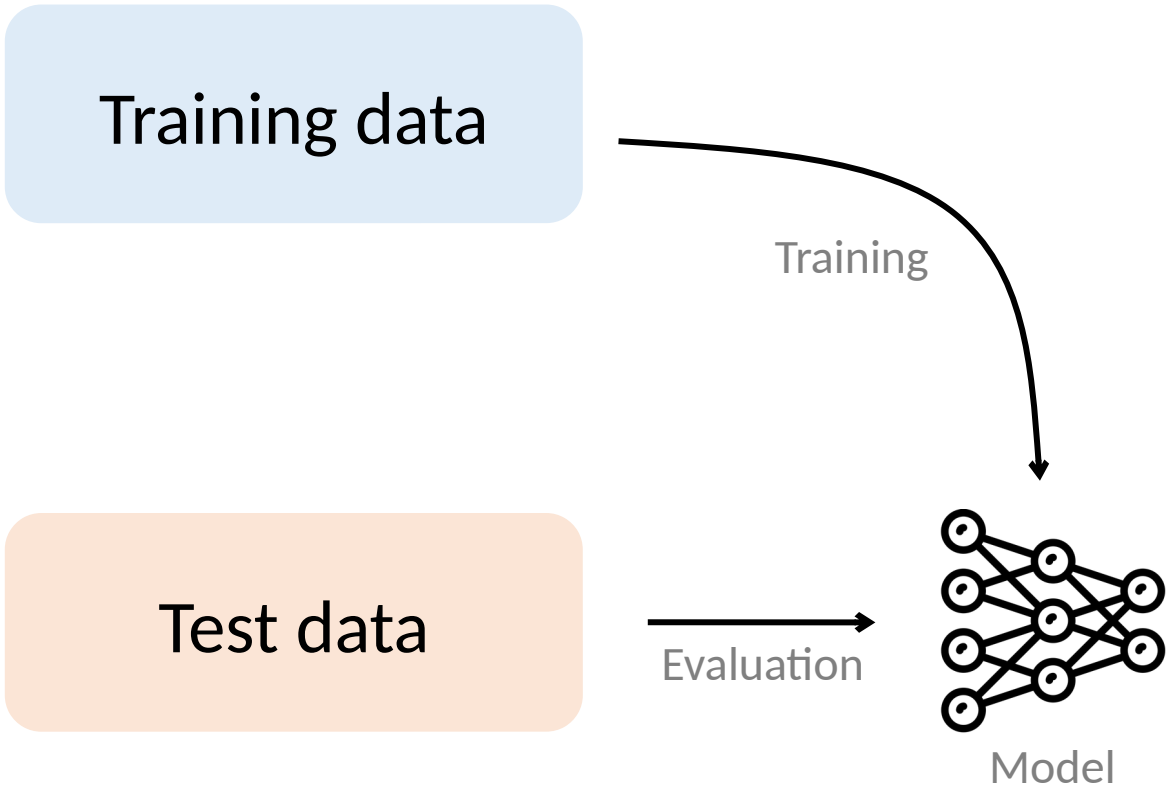
Understanding models via their training data

Understanding black-box predictions via influence functions.
Koh and Liang, ICML 2017.

Slides by Pang Wei Koh

Prior work: Focus on test data



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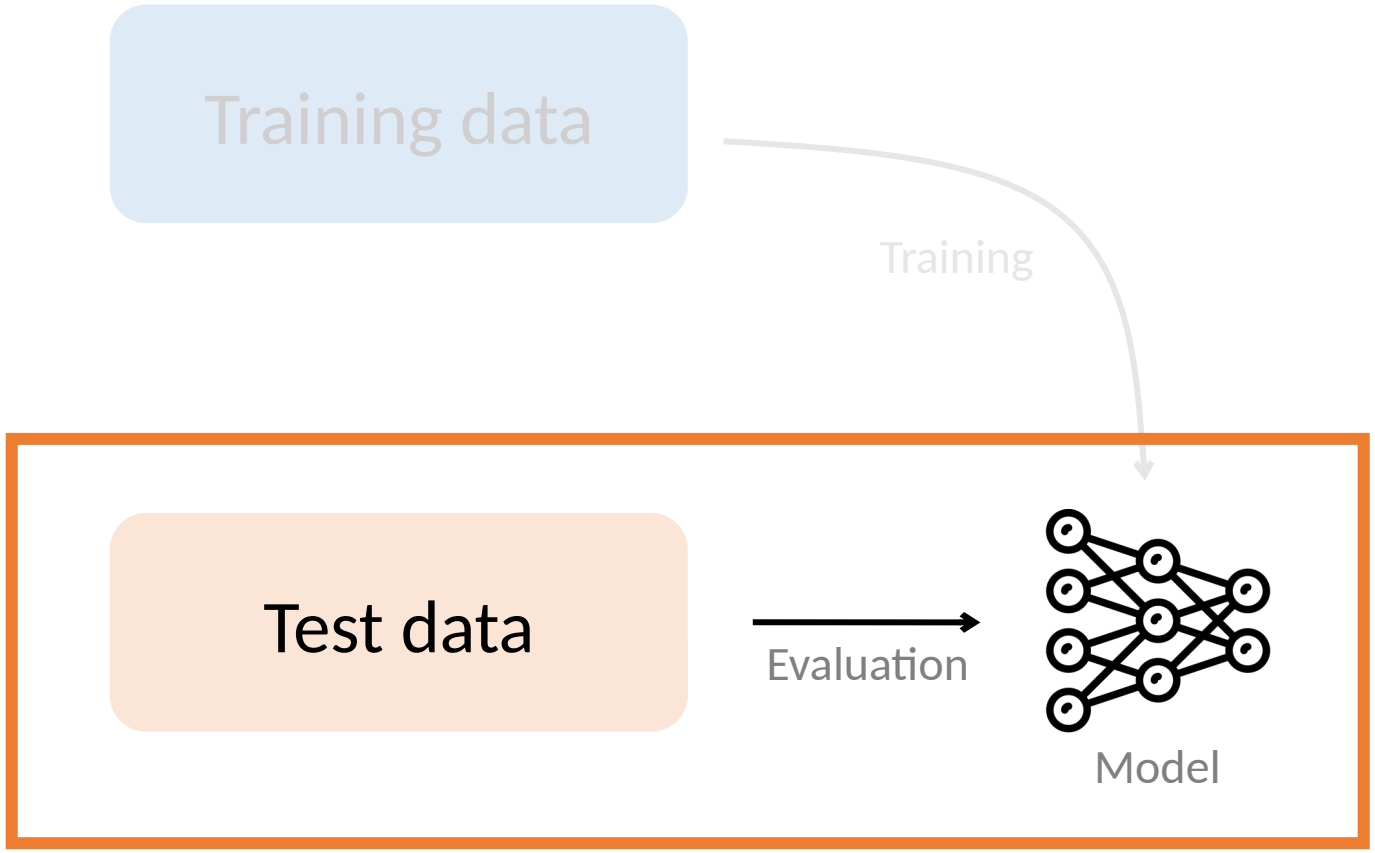
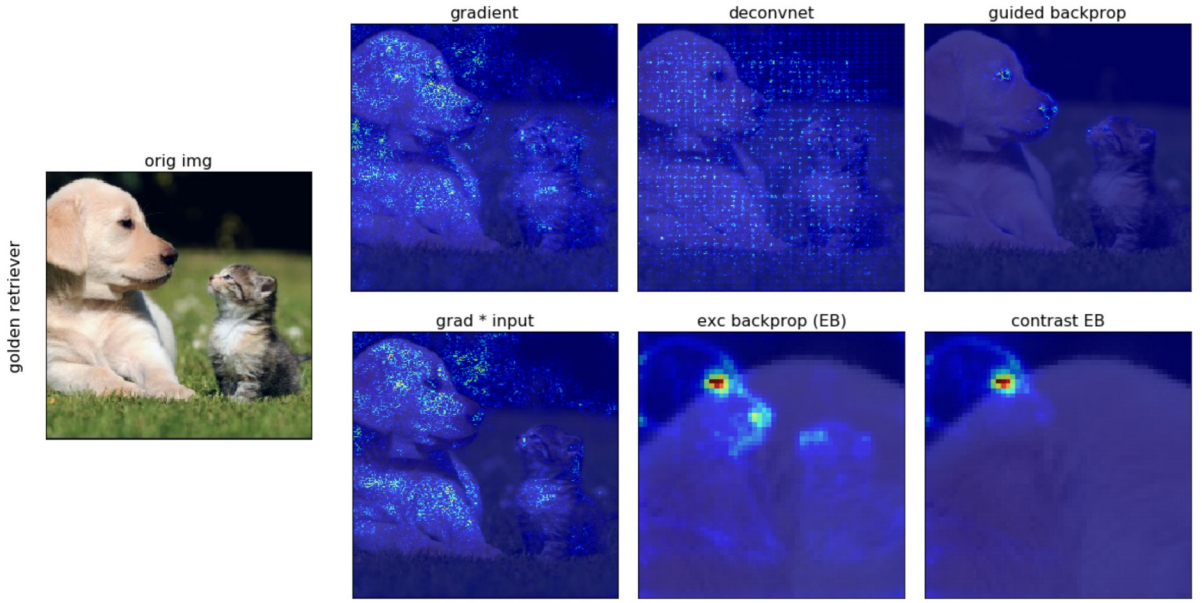


Which parts of the test input most affect the model's prediction?



[Ruth Fong's interpretability tutorial, 2019]

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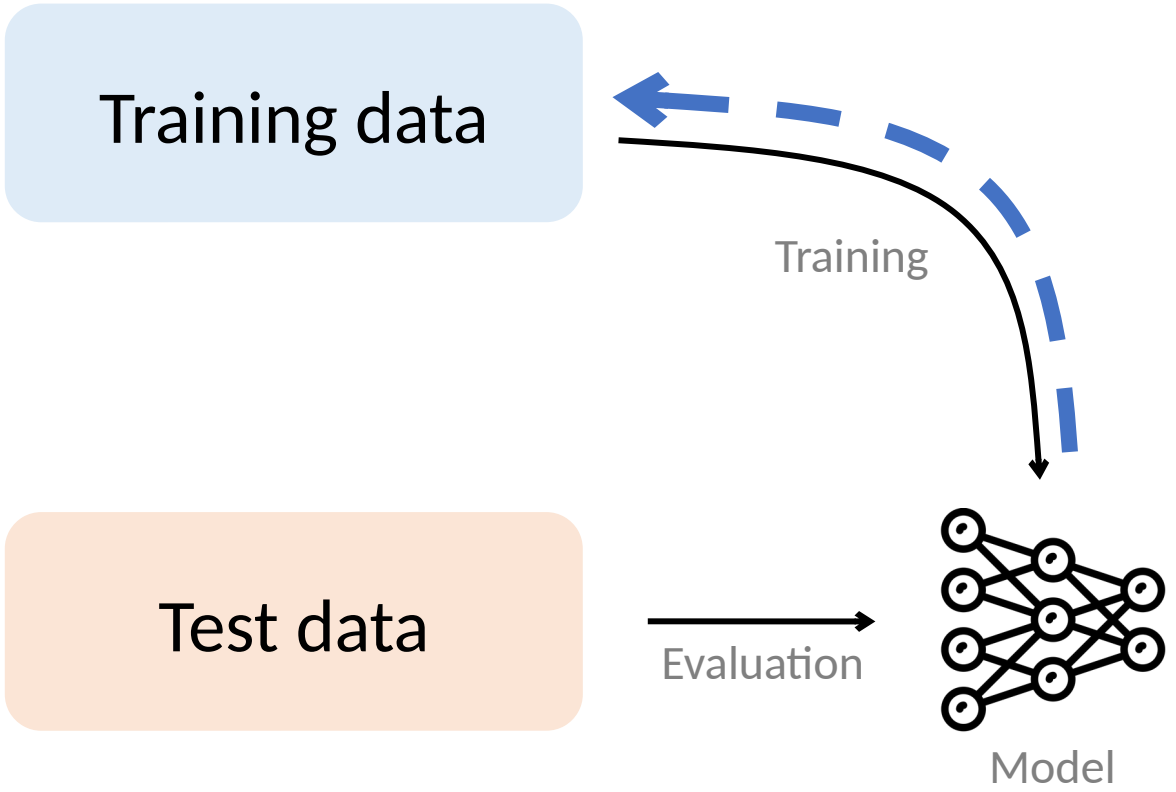
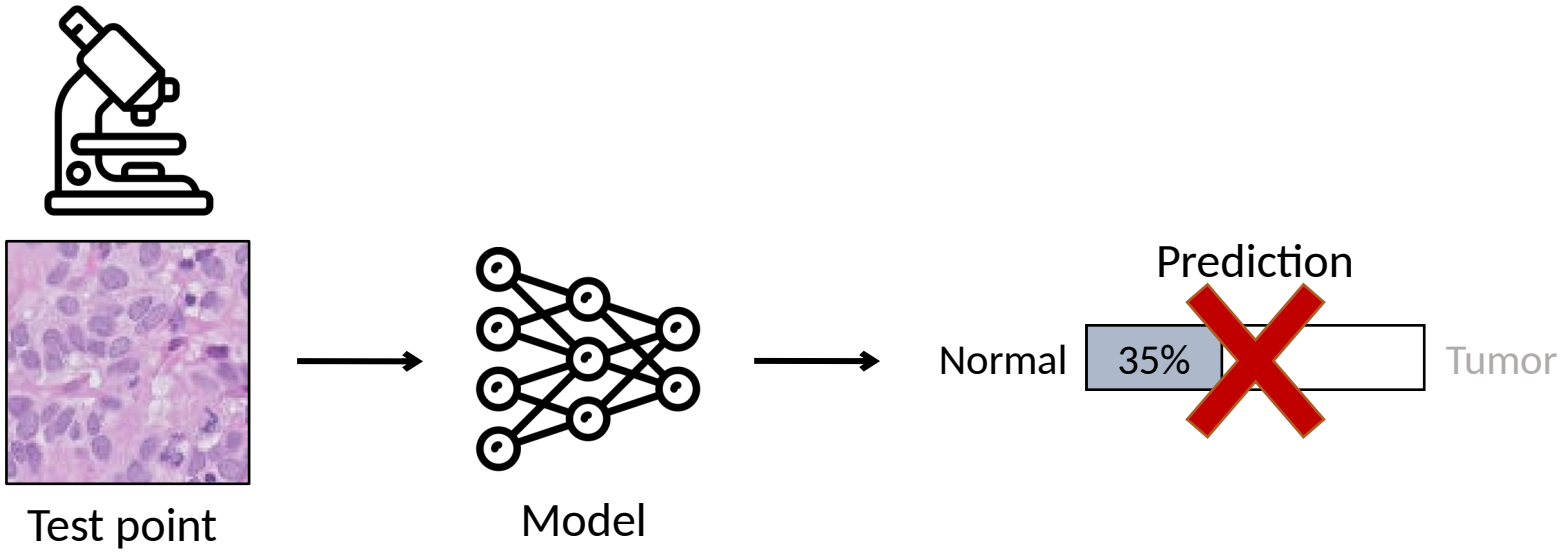
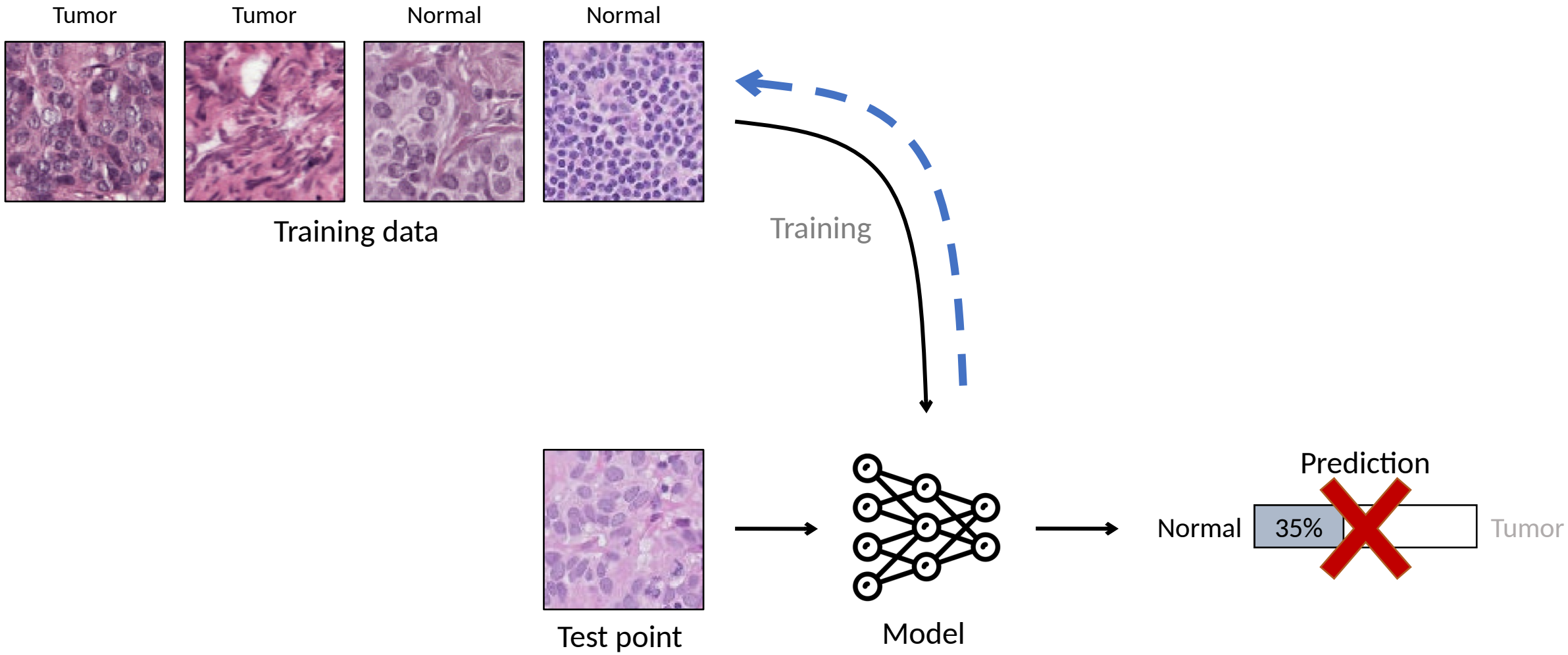
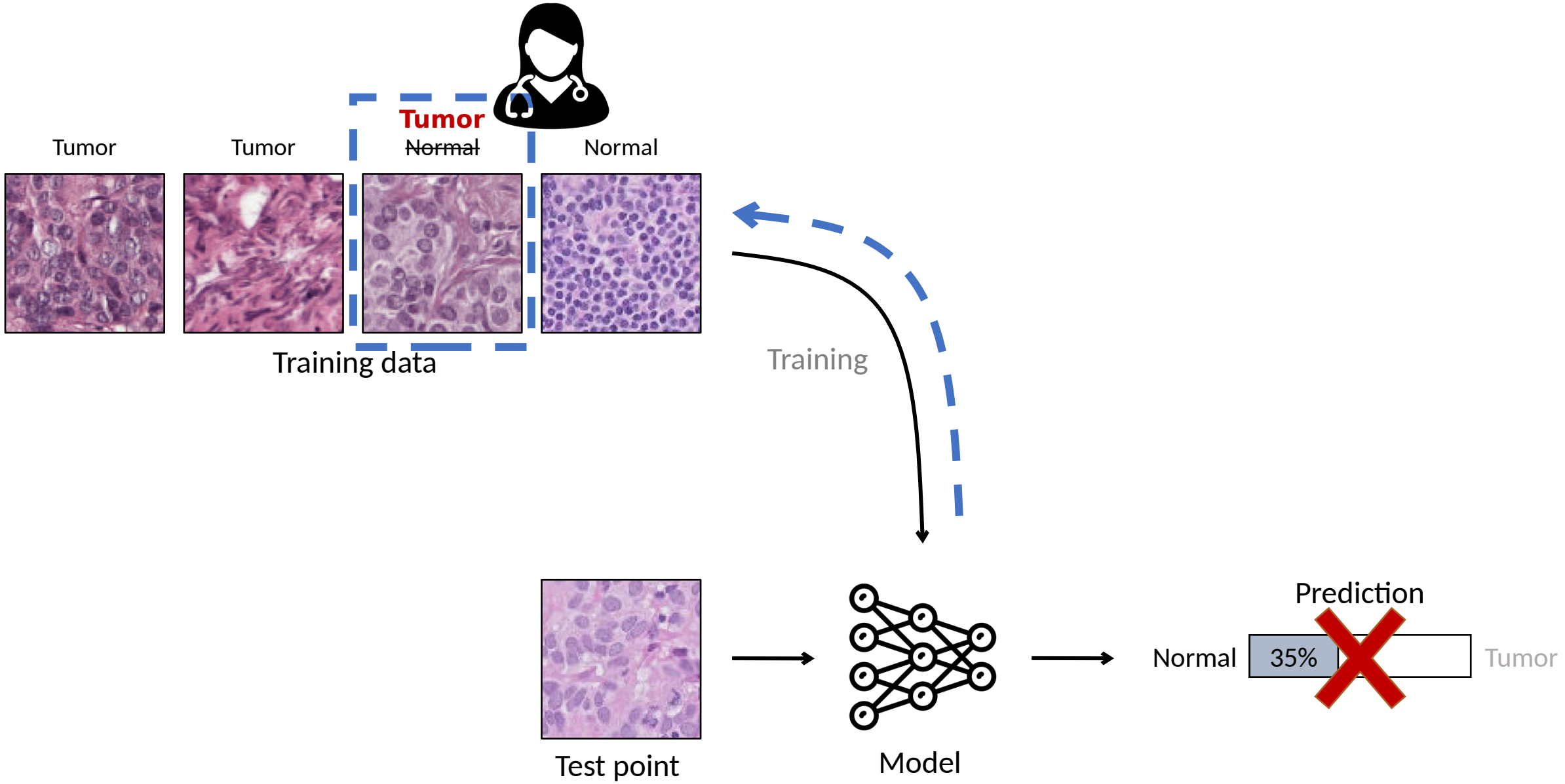
Our work: Link model to training data

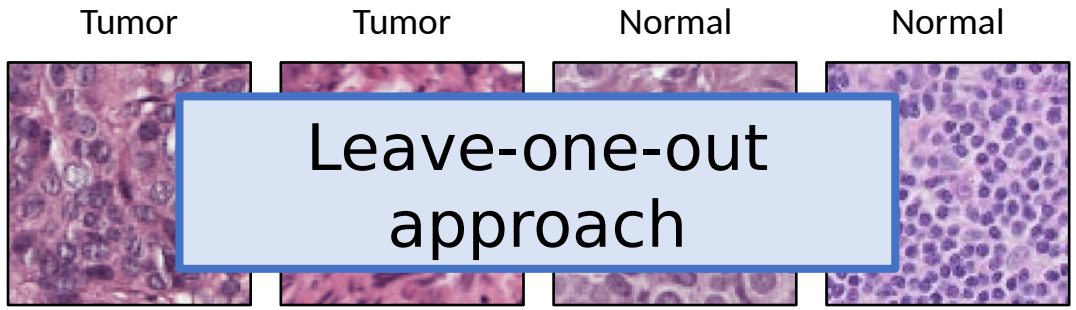


Example: Dataset debugging




Tumor
Tumor
Normal
Normal
Training data
Training
Test point
Model
Prediction
Normal
35%
Tumor

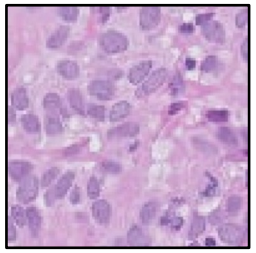

Tumor
Tumor
Tumor
~~Normal~~
Normal
Training data
Training
Test point
Model
Prediction
Normal
35%
Tumor



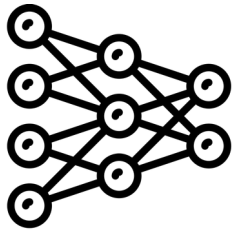
Training data

[Quenouille,1956; Tukey, 1958]

Training



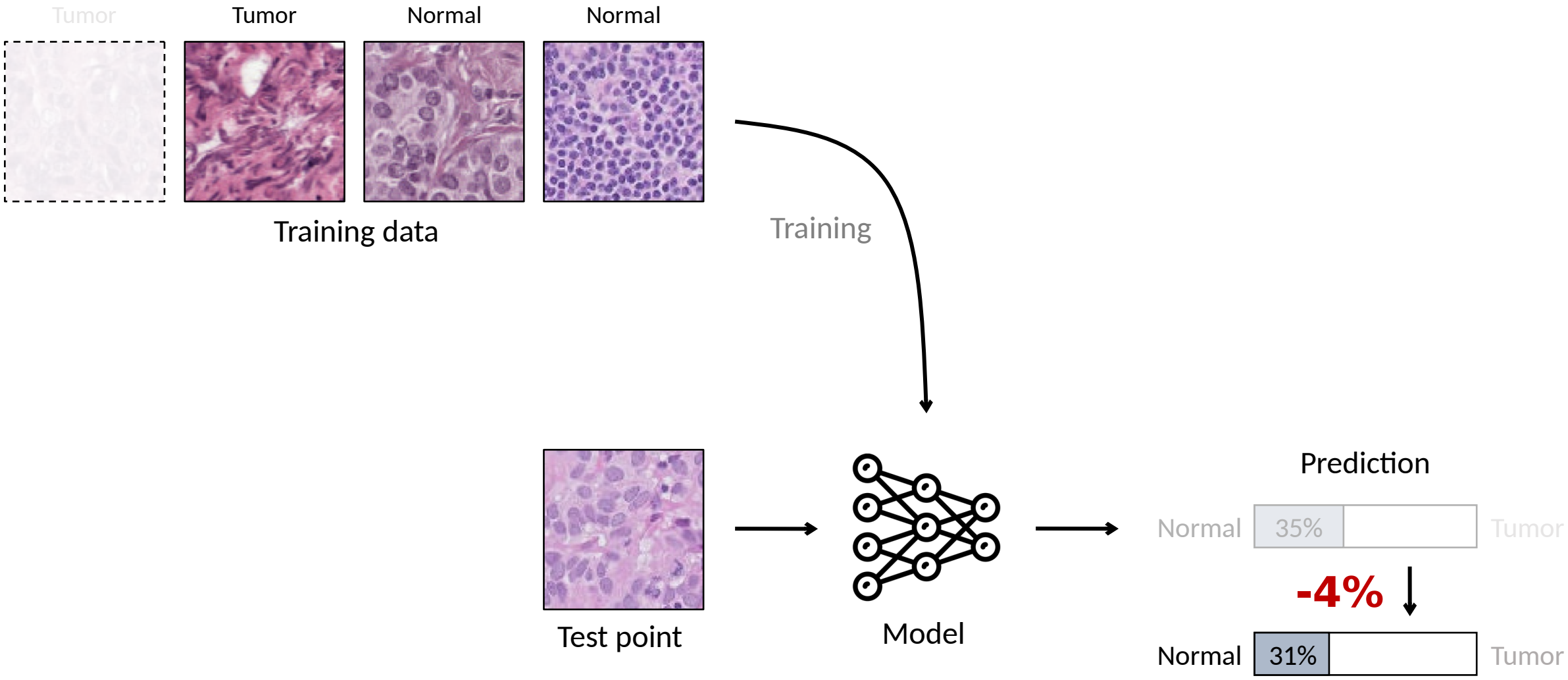
Test point

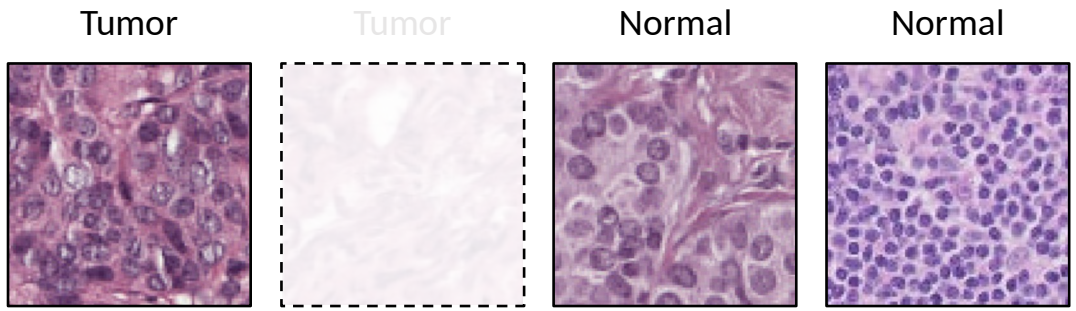


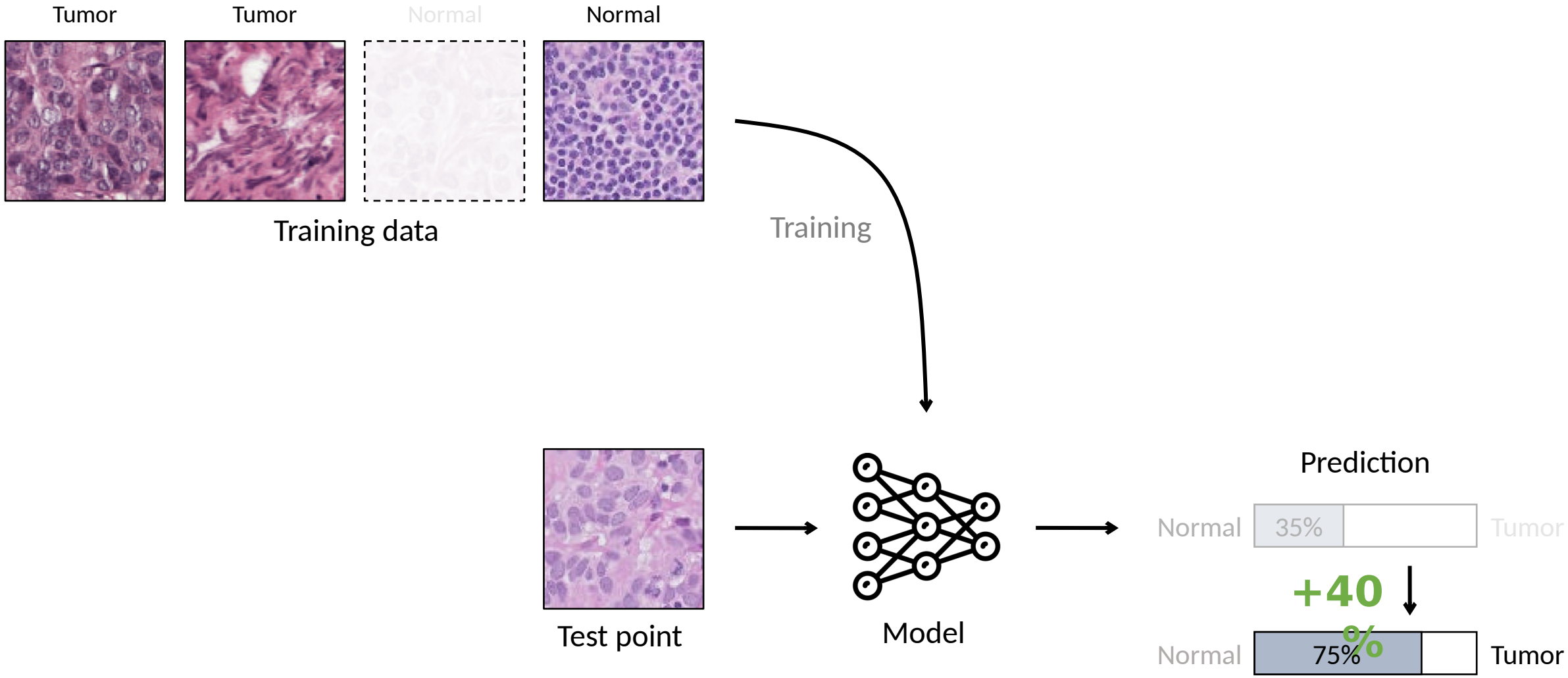
Model

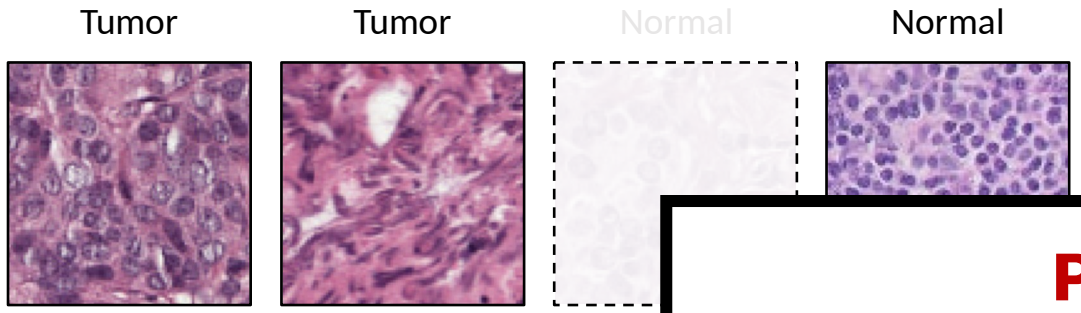
Prediction

Normal 35% Tumor


Tumor
Tumor
Normal
Normal
Training data
Training
Test point
Model
Prediction
Normal
35%
Tumor
-4%
Normal
31%
Tumor


Tumor
Tumor
Normal
Normal
Training data
Training
Test point
Model
Prediction
Normal
35%
Tumor
-2%
Normal
33%
Tumor


Tumor
Tumor
Normal
Normal
Training data
Training
Test point
Model
Prediction
Normal
35%
Tumor
+40 %
Normal
75%
Tumor

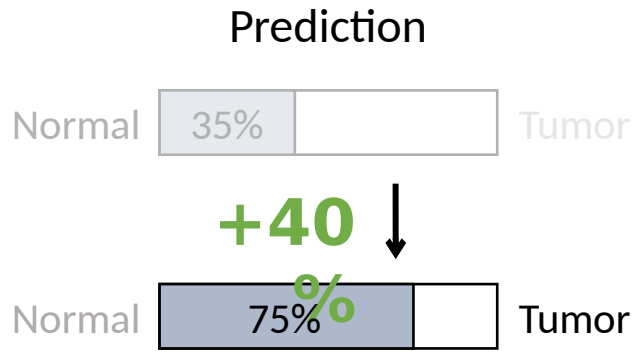


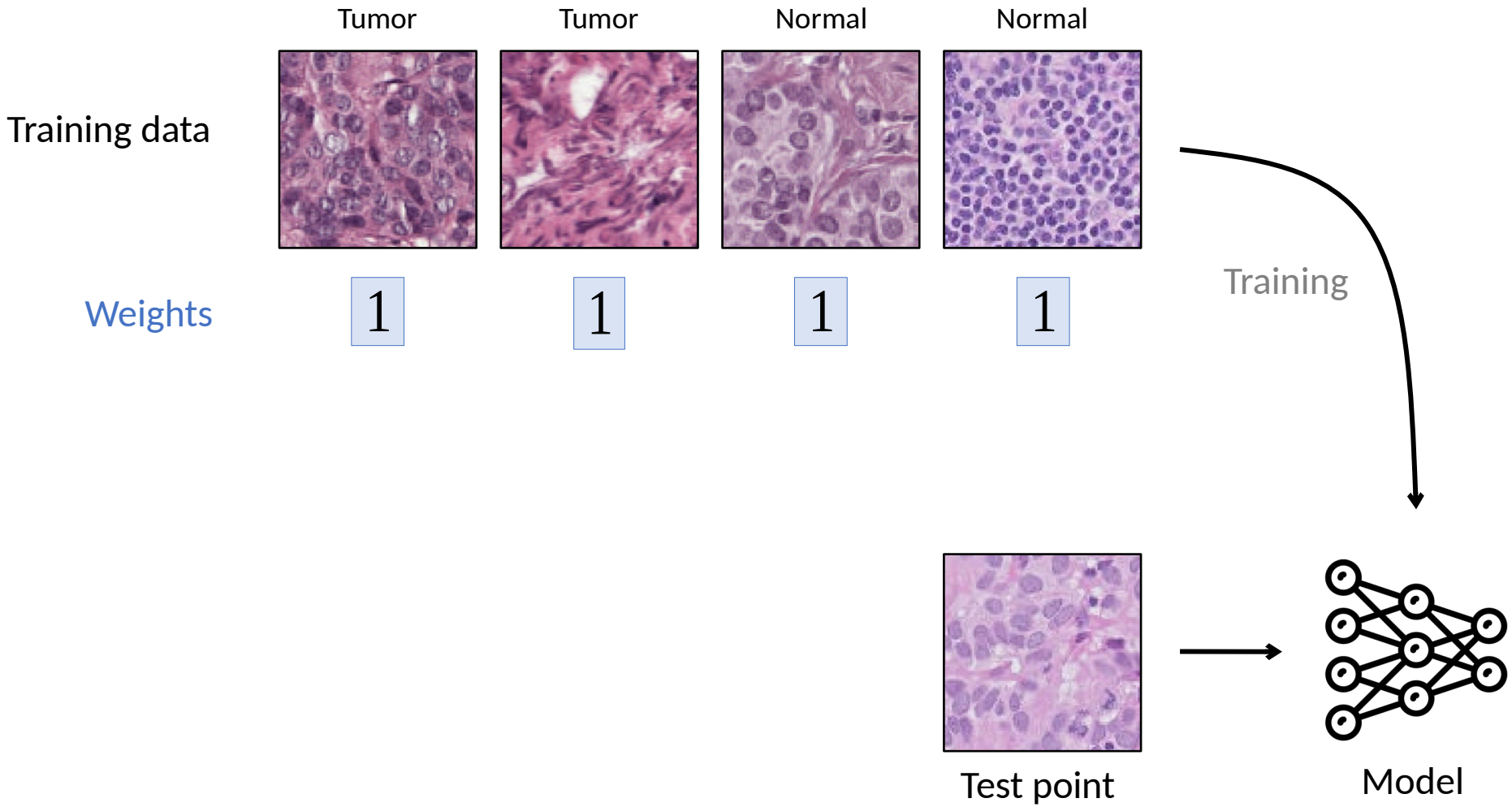
Problem

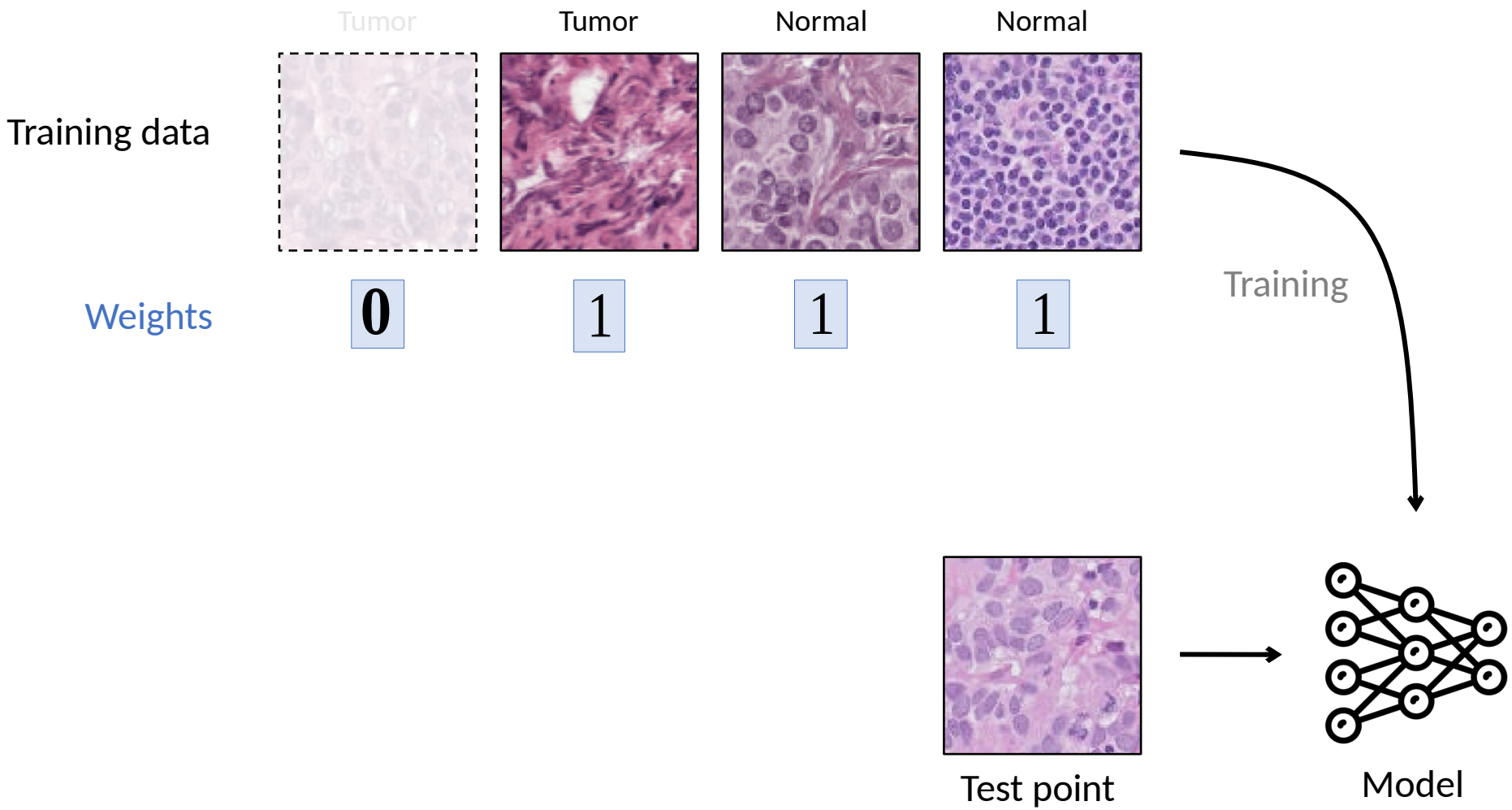
Repeatedly removing training points and retraining is too slow

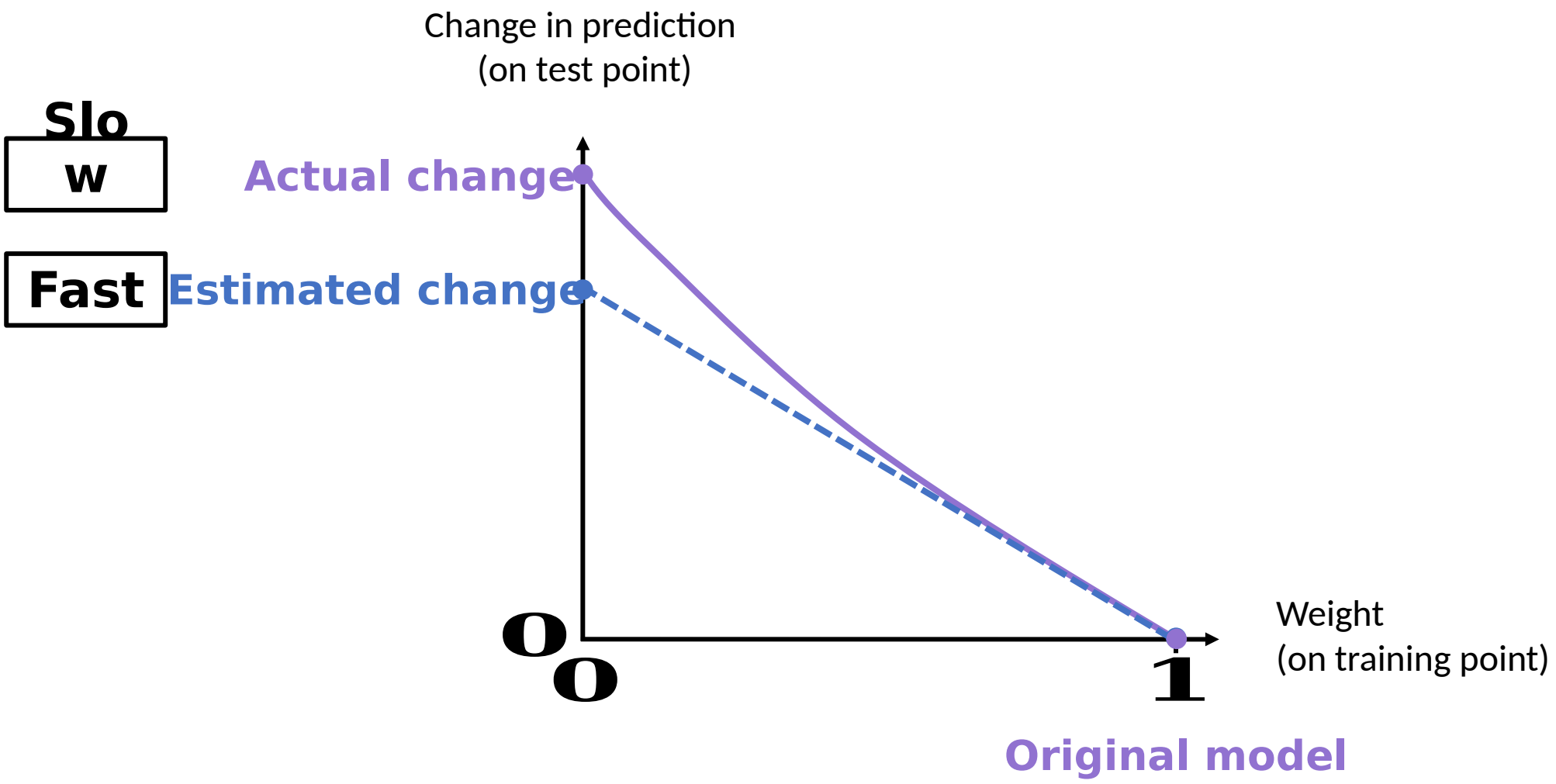
Solution

First-order Taylor approximation via influence functions




Tumor
Tumor
Normal
Normal
Training data
Weights
1
1
1
1
Training
Test point
Model


Tumor
Tumor
Normal
Normal
Training data
Weights
0
1
1
1
Training
Test point
Model


Change in prediction
(on test point)
Slo
w
Fast
Actual change
Estimated change
0
0
1
Weight
(on training point)
Original model

The influence function approximation

$$z_{test}, \hat{\theta}) \approx$$

Loss on (original)

The influence function approximation

Change in loss on after removing

$$\ell\left(z_{test}, \hat{\theta}_{-z_{train}}\right) - \ell\left(z_{test}, \hat{\theta}\right) \approx$$

Loss on (after removing)

Loss on (original)

The influence function approximation

Change in loss on after removing

$$\ell\left(z_{test}, \hat{\theta}_{-z_{train}}\right) - \ell\left(z_{test}, \hat{\theta}\right) \approx$$

The influence function approximation

Change in loss on after removing

$$\ell\left(z_{test}, \hat{\theta}_{-z_{train}}\right) - \ell\left(z_{test}, \hat{\theta}\right) \approx \nabla_\theta \ell\left(z_{test}, \hat{\theta}\right)^T H_{\hat{\theta}}^{-1} \nabla_\theta \ell\left(z_{train}, \hat{\theta}\right)$$

Gradient of loss on

Inverse of the Hessian

Gradient of loss on

The influence function approximation

Doesn't require retraining!

Change in loss on after removing

$$\ell\left(z_{test}, \hat{\theta}_{-z_{train}}\right) - \ell\left(z_{test}, \hat{\theta}\right) \approx \nabla_\theta \ell\left(z_{test}, \hat{\theta}\right)^T H_{\hat{\theta}}^{-1} \nabla_\theta \ell\left(z_{train}, \hat{\theta}\right)$$

Gradient of loss on

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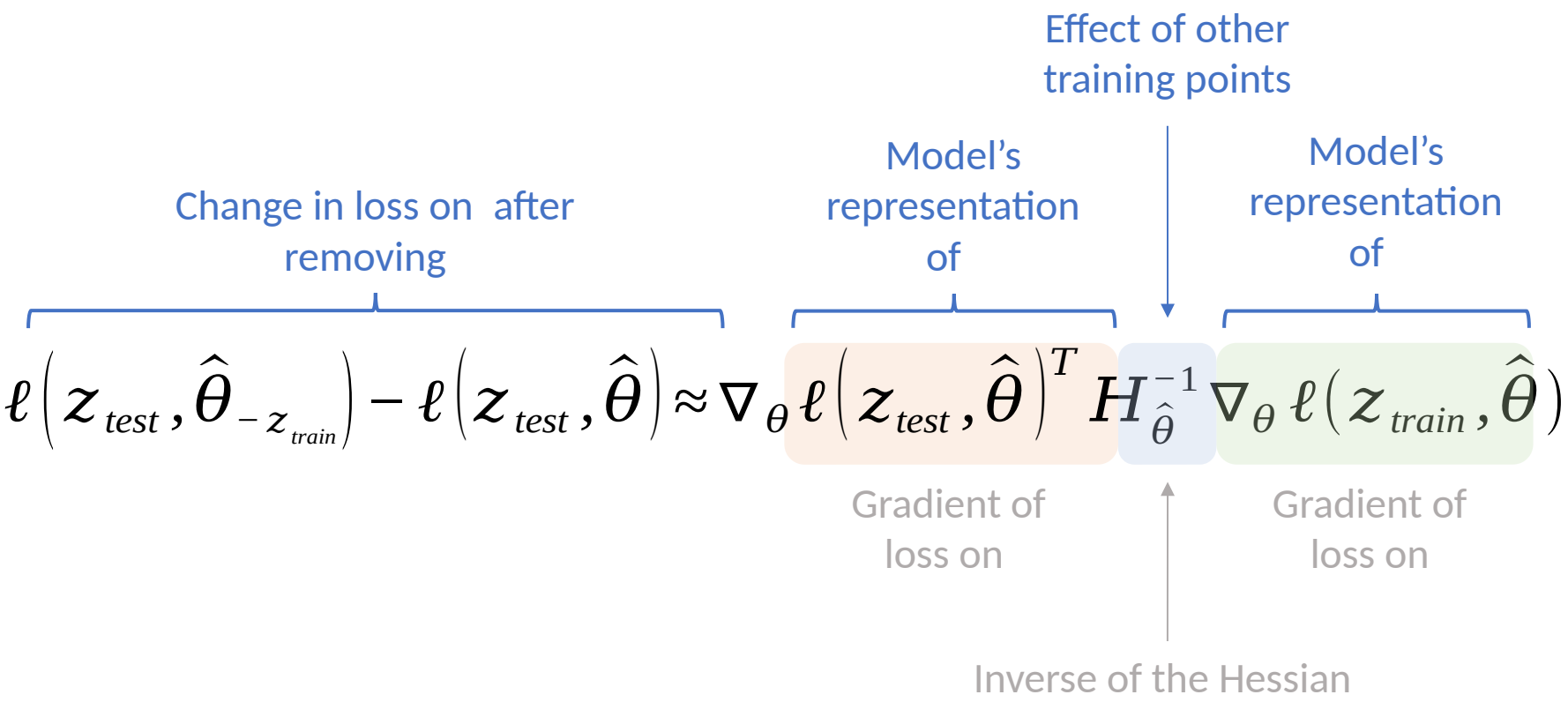
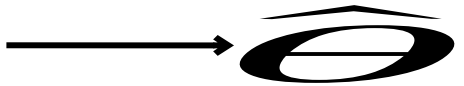
Gradient of loss on

The influence function approximation



A little bit more technical details ...

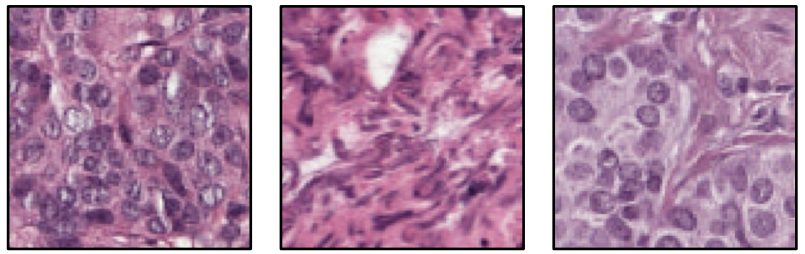
Training data



$\ell\left(z_{test}, \hat{\theta}\right)$

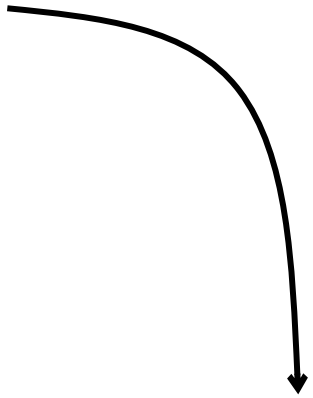
Training data

Tumor | Tumor | Normal



Weights

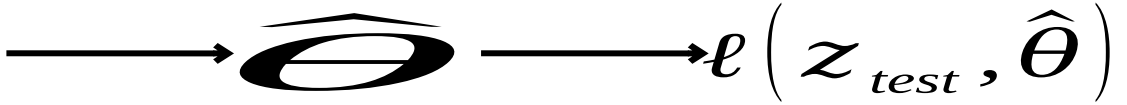
$+\epsilon$ | 1 | 1



$$\hat{\theta}_{\epsilon,z} \stackrel{\text{def}}{=} \arg\min_{\theta\in\Theta} \frac{1}{n}\sum_{i=1}^{n} L(z_i,\theta) + \epsilon L(z,\theta)$$

A little bit more technical details …

Training data



$$\left.\frac{d\hat{\theta}_{\epsilon,z}}{d\epsilon}\right|_{\epsilon=0} \qquad \left.\frac{dL(z_{\text{test}}, \hat{\theta}_{\epsilon,z})}{d\epsilon}\right|_{\epsilon=0}$$

$$= \nabla_\theta L(z_{\text{test}}, \hat{\theta})^\top \left.\frac{d\hat{\theta}_{\epsilon,z}}{d\epsilon}\right|_{\epsilon=0}$$

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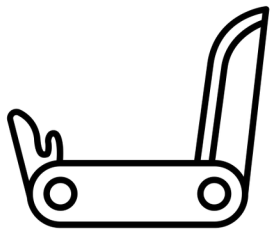
From classical to modern settings



Jaeckel, 1972. The infinitesimal jackknife.

Hampel, 1974. The influence curve and its role in robust estimation.

Cook, 1977. Detection of influential observations in linear regression.

…

From classical to modern settings

Small datasets
Low-dimensional

Large datasets
High-dimensional

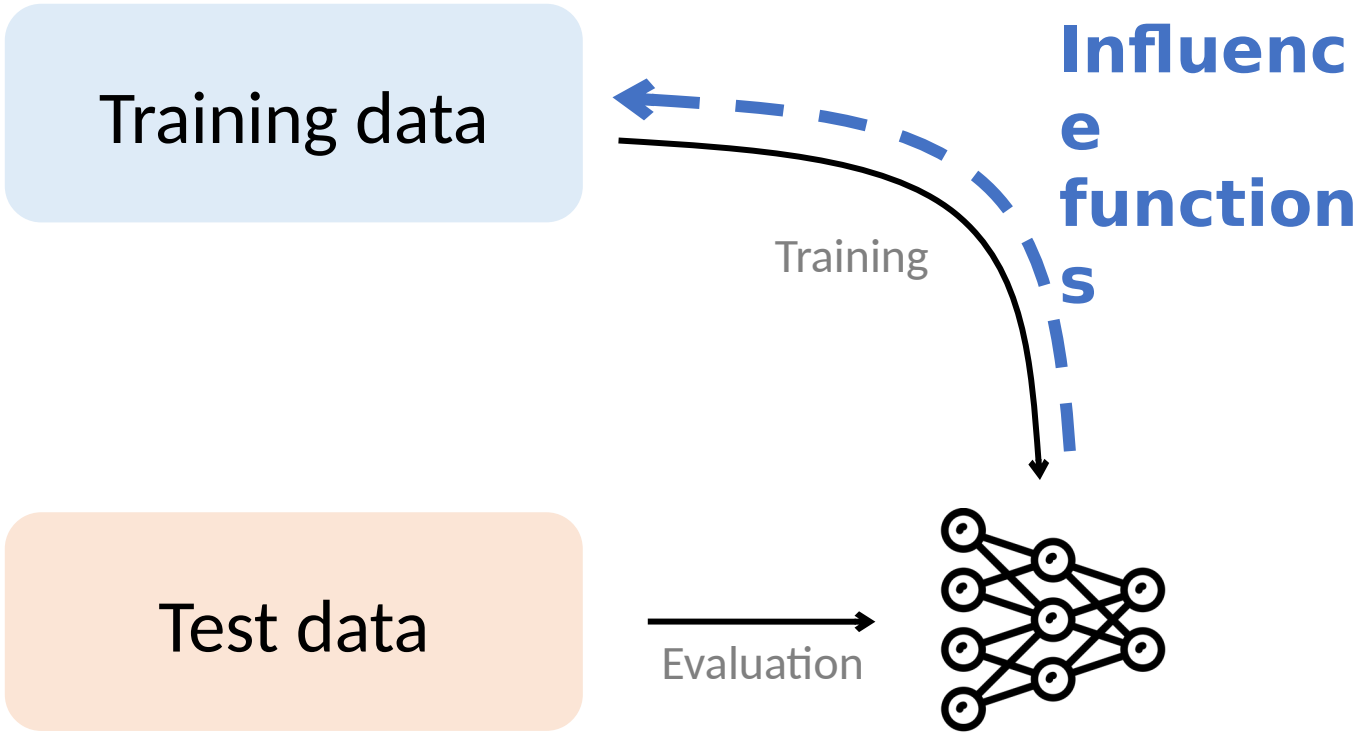
Difficult to compute

$$\nabla_\theta \ell\left(z_{test}, \hat{\theta}\right)^T \boldsymbol{H}_{\hat{\boldsymbol{\theta}}}^{-1} \nabla_\theta \ell\left(z_{train}, \hat{\theta}\right)$$

We use tools from
2nd-order optimization & stochastic estimation
[Pearlmutter, 1994; Martens, 2010, Agarwal et al., 2017]

Removing single points

Understanding Models via Their Training Data



Where can you apply influence functions?

Impact

Robustness

Training set biases
[Ren et al., 2018]

Inference reliability
[Broderick et al., 2021]

Cross-validation
[Stephenson et al., 2020]

Memorization
[Feldman, 2019]

Applications

Data distillation
[Wang et al., 2020]

Data valuation
[Jia et al., 2019]

Active learning
[Gudovskiy et al., 2020]

Data debugging
[Guo et al., 2021]

Fairness & security

Algorithmic bias
[Verma et al., 2021]

Data labor
[Arrieta-Ibarra, 2018]

Privacy
[Shokri et al., 2021]

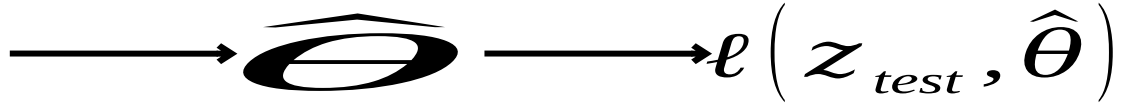
Data poisoning
[Chen et al., 2017]

Limitations

- What assumptions on the models are needed to calculate influence functions?
- Can you calculate influence functions for a neural network with the recipe described earlier?

$$\left.\frac{dL(z_{\text{test}}, \hat{\theta}_{\epsilon,z})}{d\epsilon}\right|_{\epsilon=0}$$

Training data



$$\hat{\theta}_{\epsilon,z} \stackrel{\text{def}}{=} \arg\min_{\theta\in\Theta} \frac{1}{n}\sum_{i=1}^{n} L(z_i,\theta) + \epsilon L(z,\theta)$$

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If Influence Functions are the Answer, Then What is the Question?

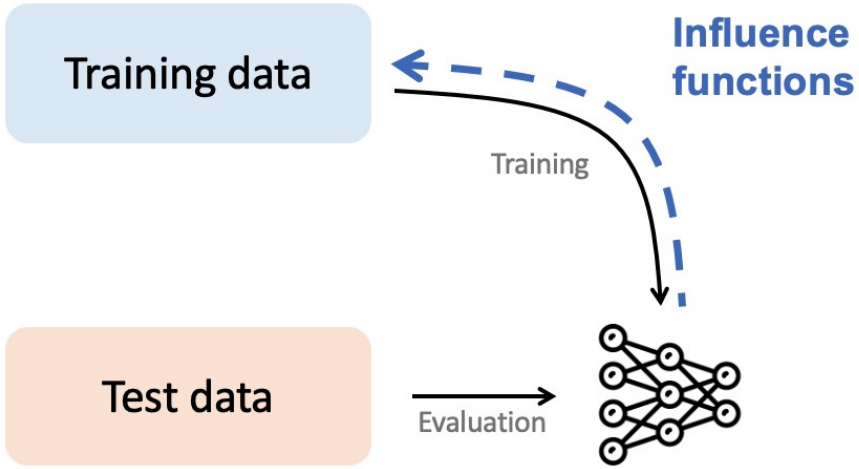
Juhan Bae*†, Nathan Ng†‡, Alston Lo†, Marzyeh Ghassemi‡, Roger Grosse†

Non-convexity Non-convergence

$$\hat{\theta}_{\epsilon,z} \overset{\text{def}}{=} \arg\min_{\theta\in\Theta} \frac{1}{n}\sum_{i=1}^{n} L(z_i,\theta) + \epsilon L(z,\theta)$$

Summary

- Link model behavior to training data
- Efficient to calculate
- Many applications



- Limitations when applying to complex models

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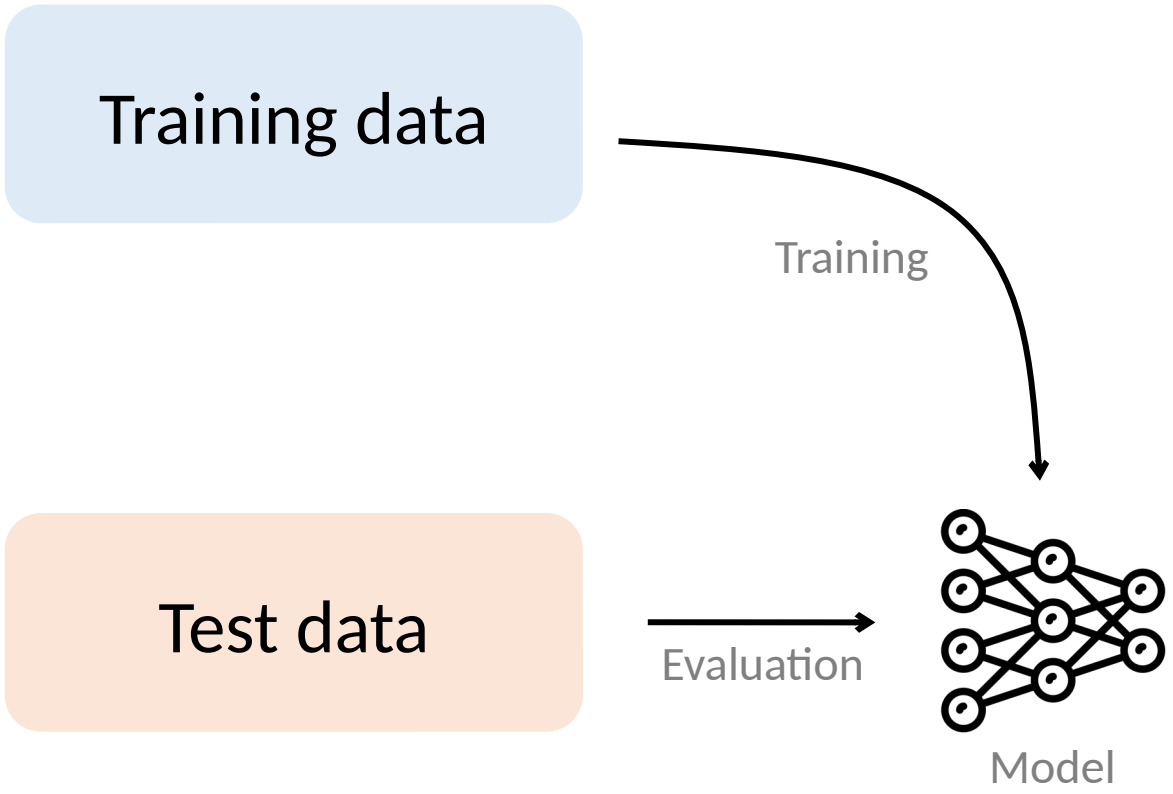

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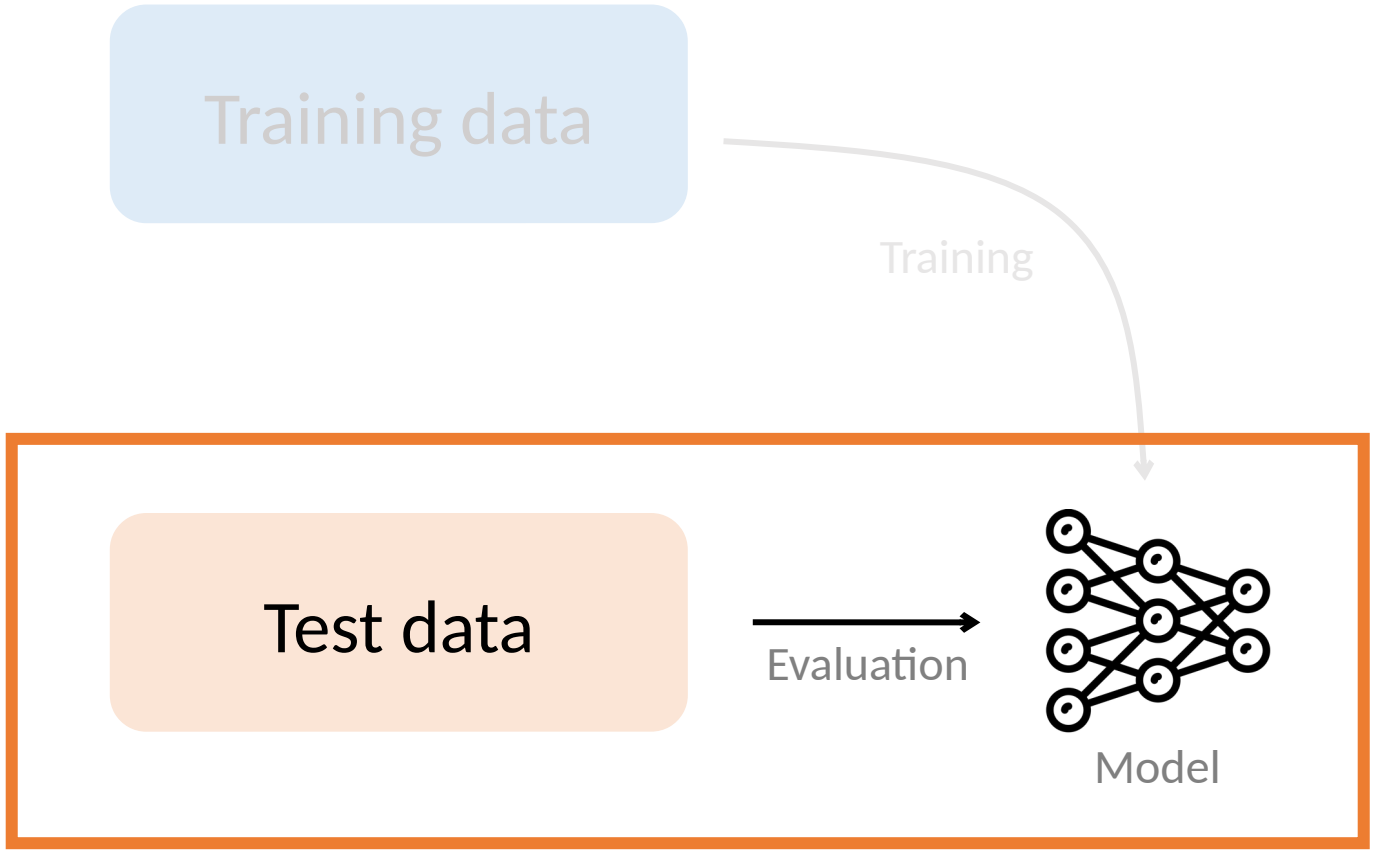

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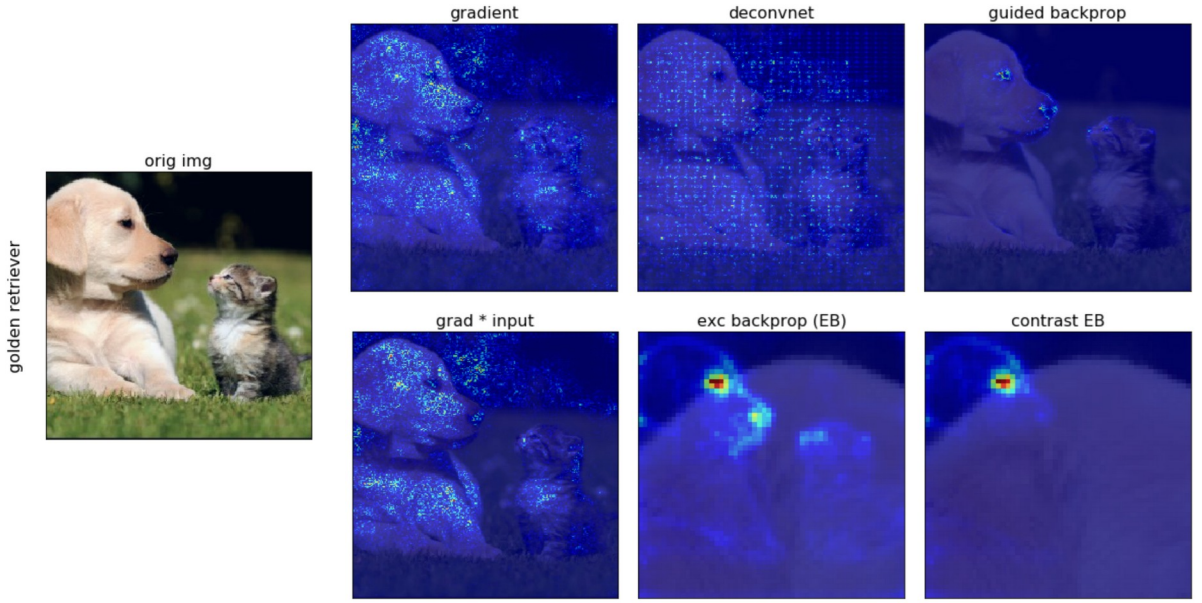

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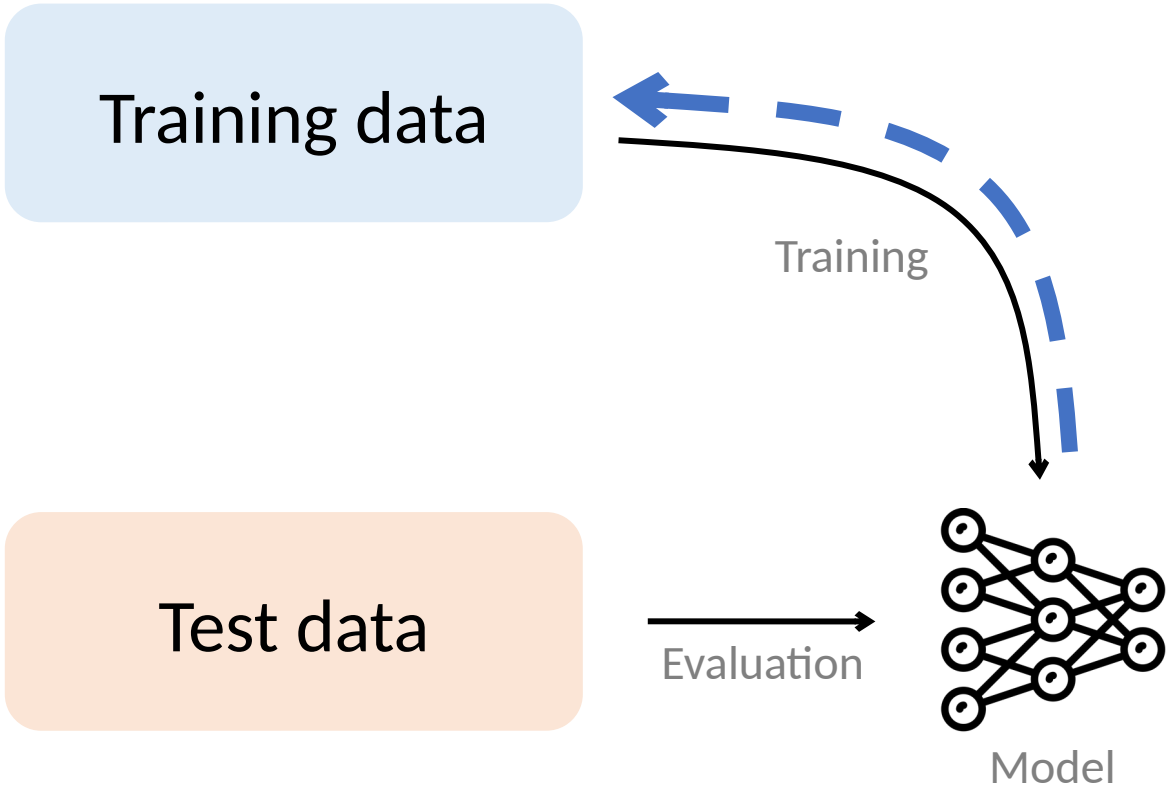

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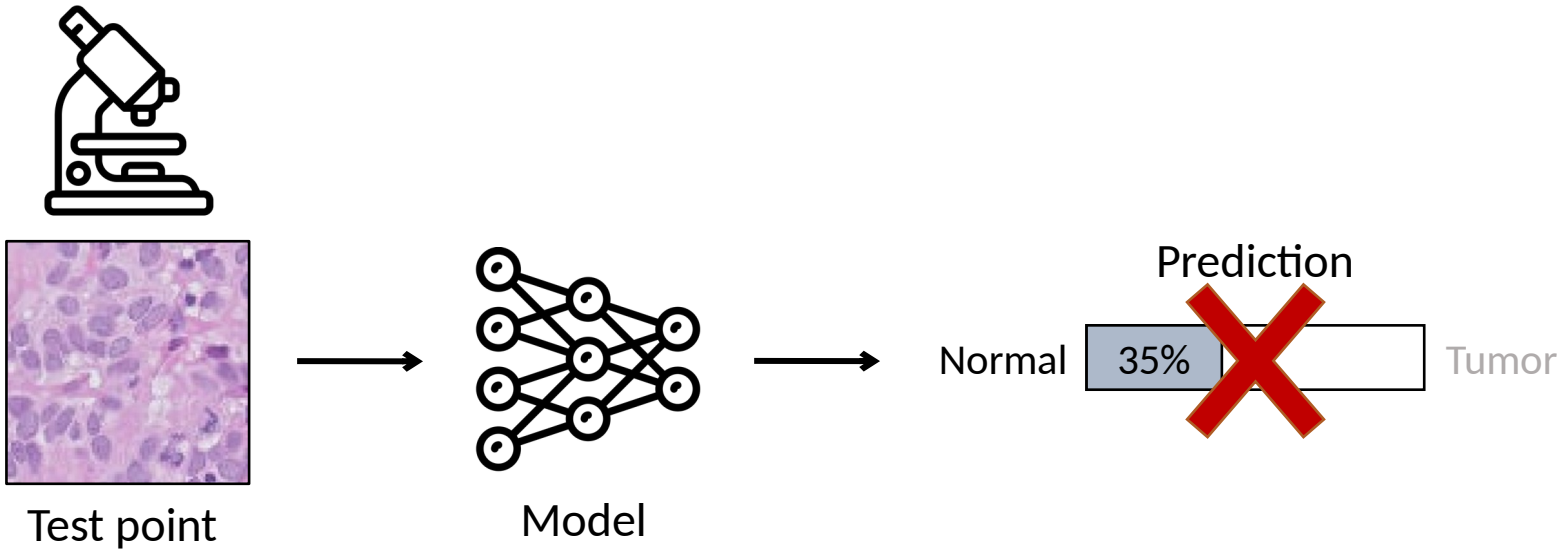
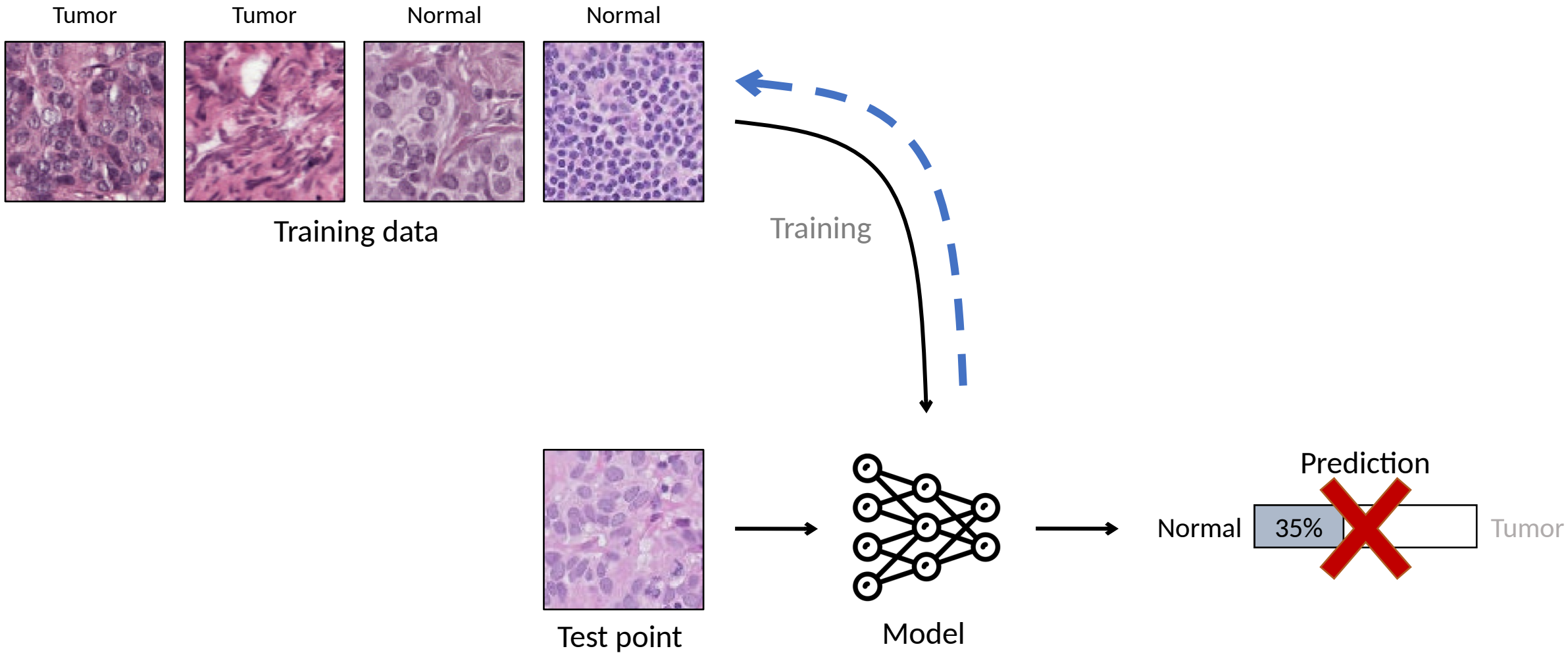
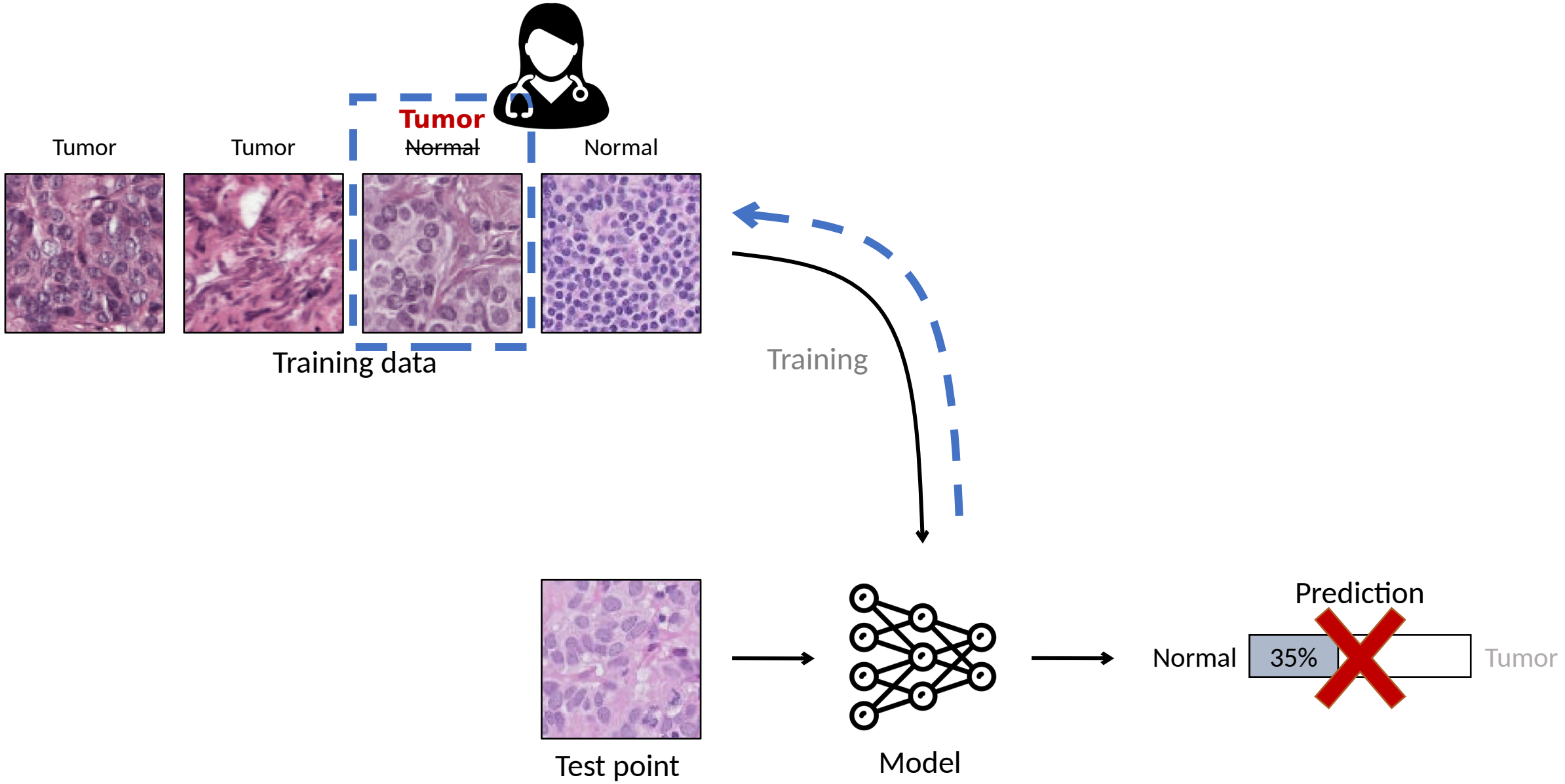
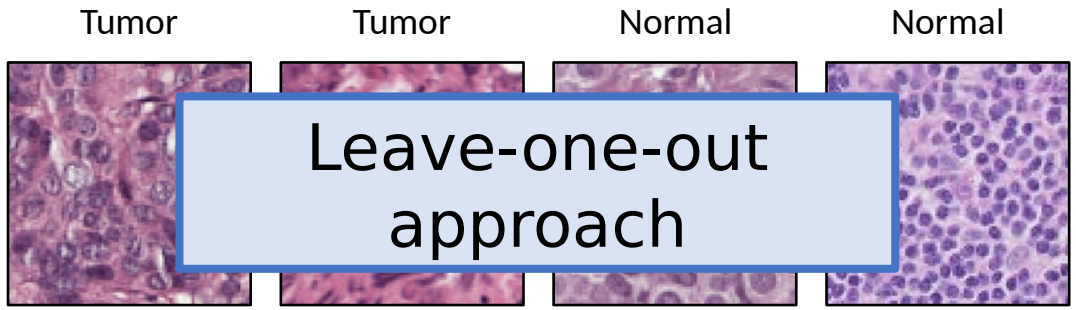



Tumor
Tumor
Normal
Normal
Training data
Training
Test point
Model
Prediction
Normal
35%
Tumor

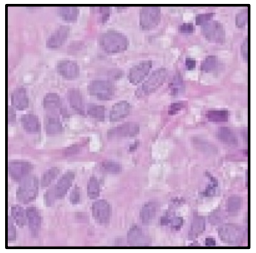

Tumor
Tumor
Tumor
~~Normal~~
Normal
Training data
Training
Test point
Model
Prediction
Normal
35%
Tumor



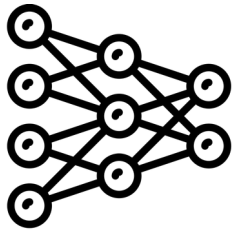
Training data

[Quenouille,1956; Tukey, 1958]

Training



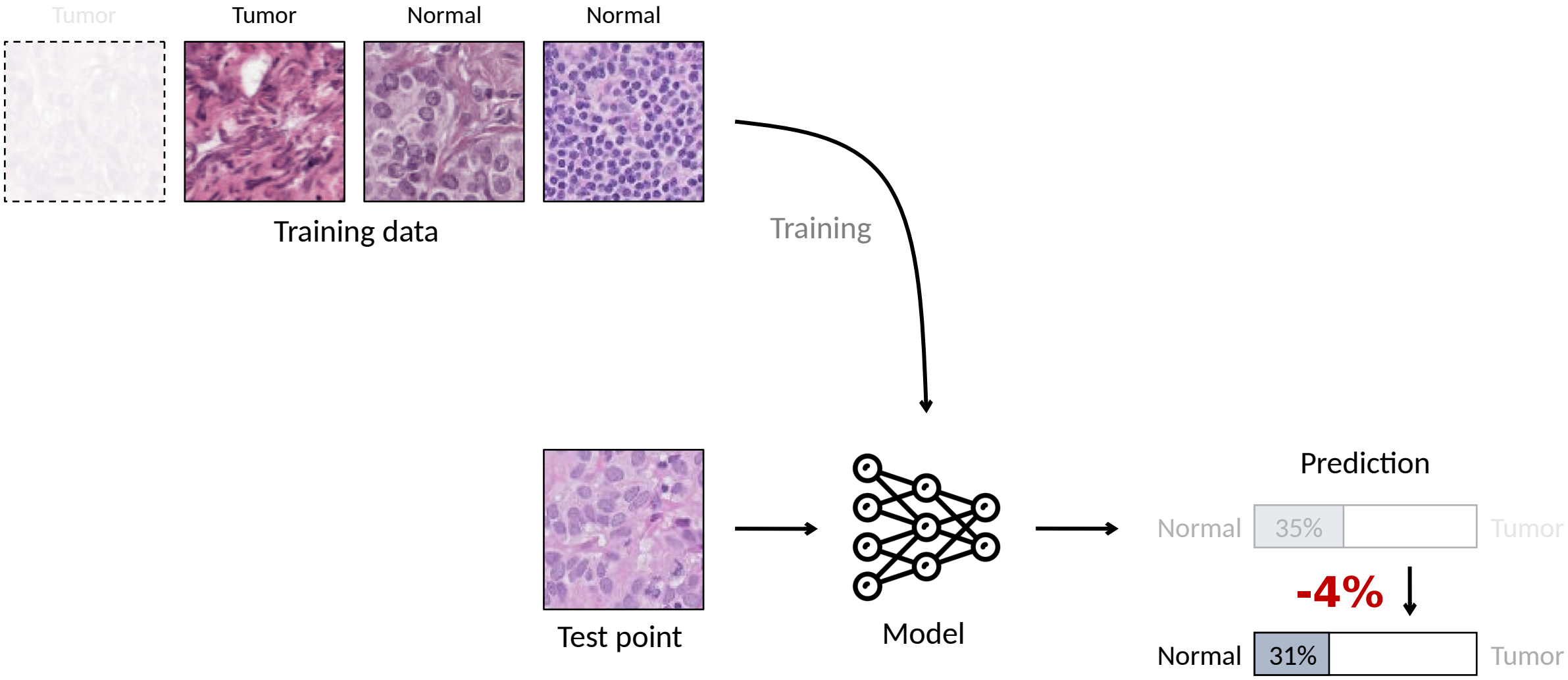
Test point

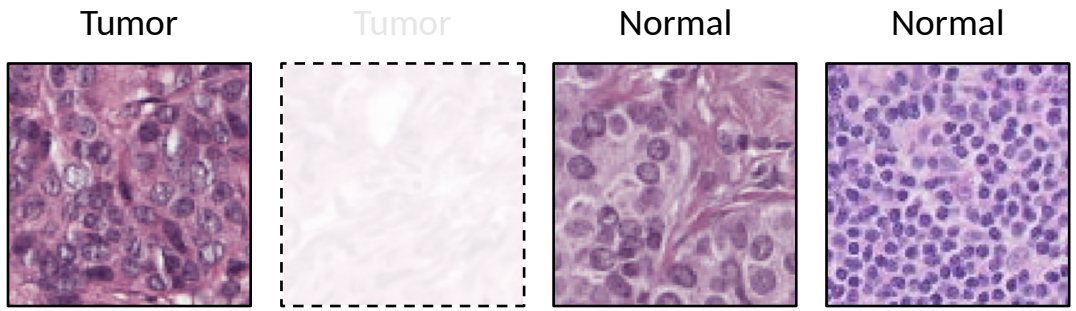


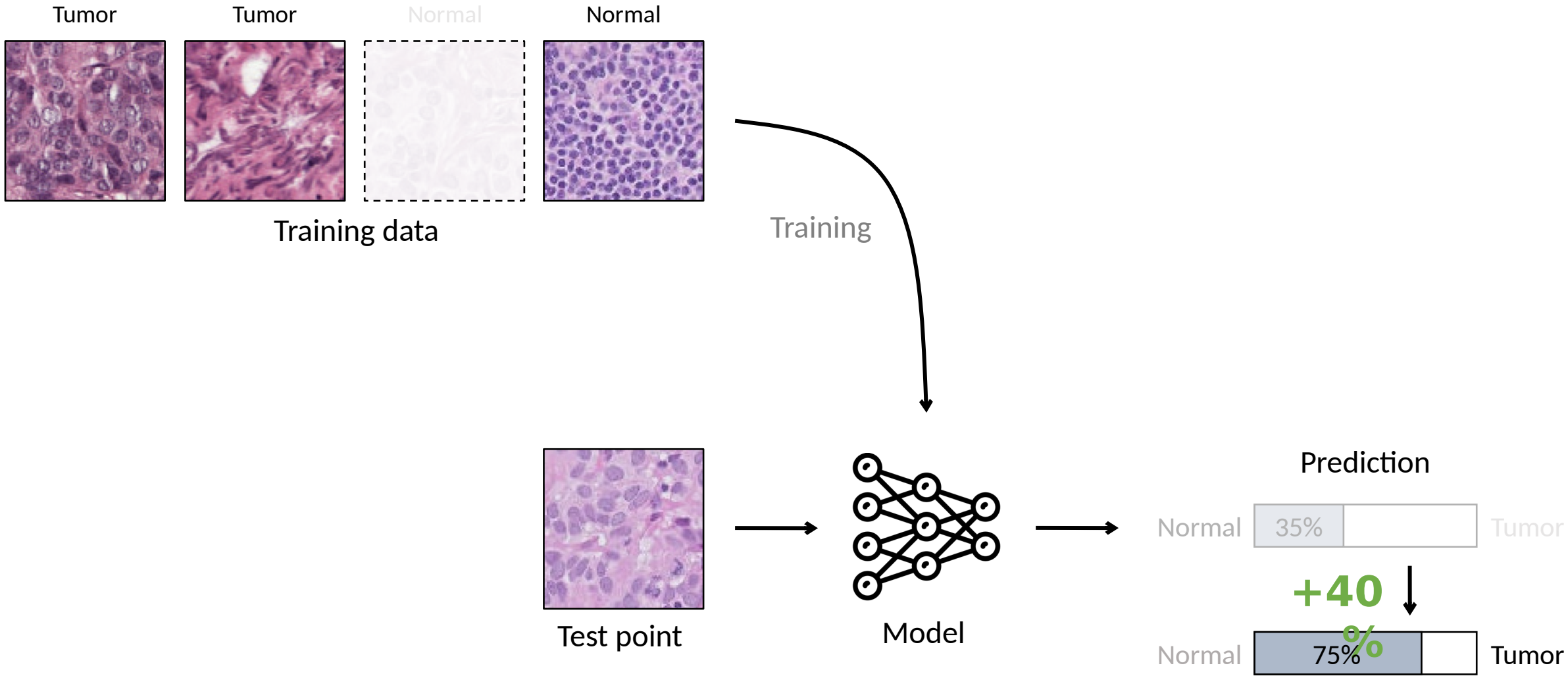
Model

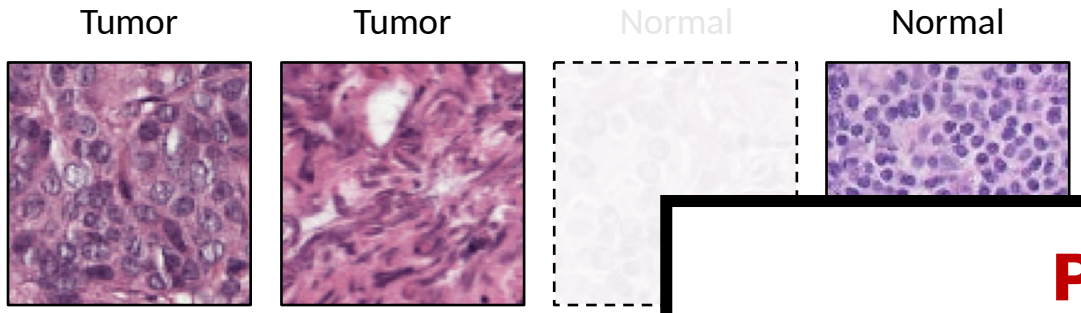
Prediction

Normal 35% Tumor


Tumor
Tumor
Normal
Normal
Training data
Training
Test point
Model
Prediction
Normal
35%
Tumor
-4%
Normal
31%
Tumor


Tumor
Tumor
Normal
Normal
Training data
Training
Test point
Model
Prediction
Normal
35%
Tumor
-2%
Normal
33%
Tumor


Tumor
Tumor
Normal
Normal
Training data
Training
Test point
Model
Prediction
Normal
35%
Tumor
+40 %
Normal
75%
Tumor

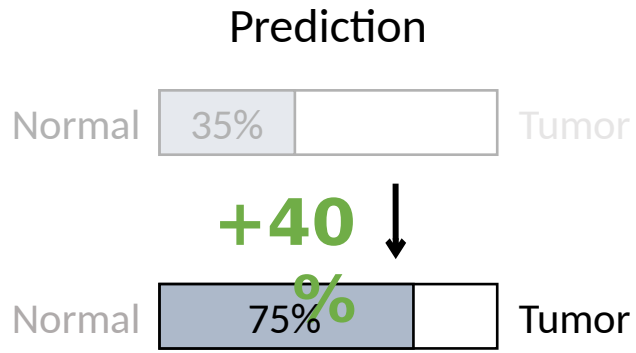


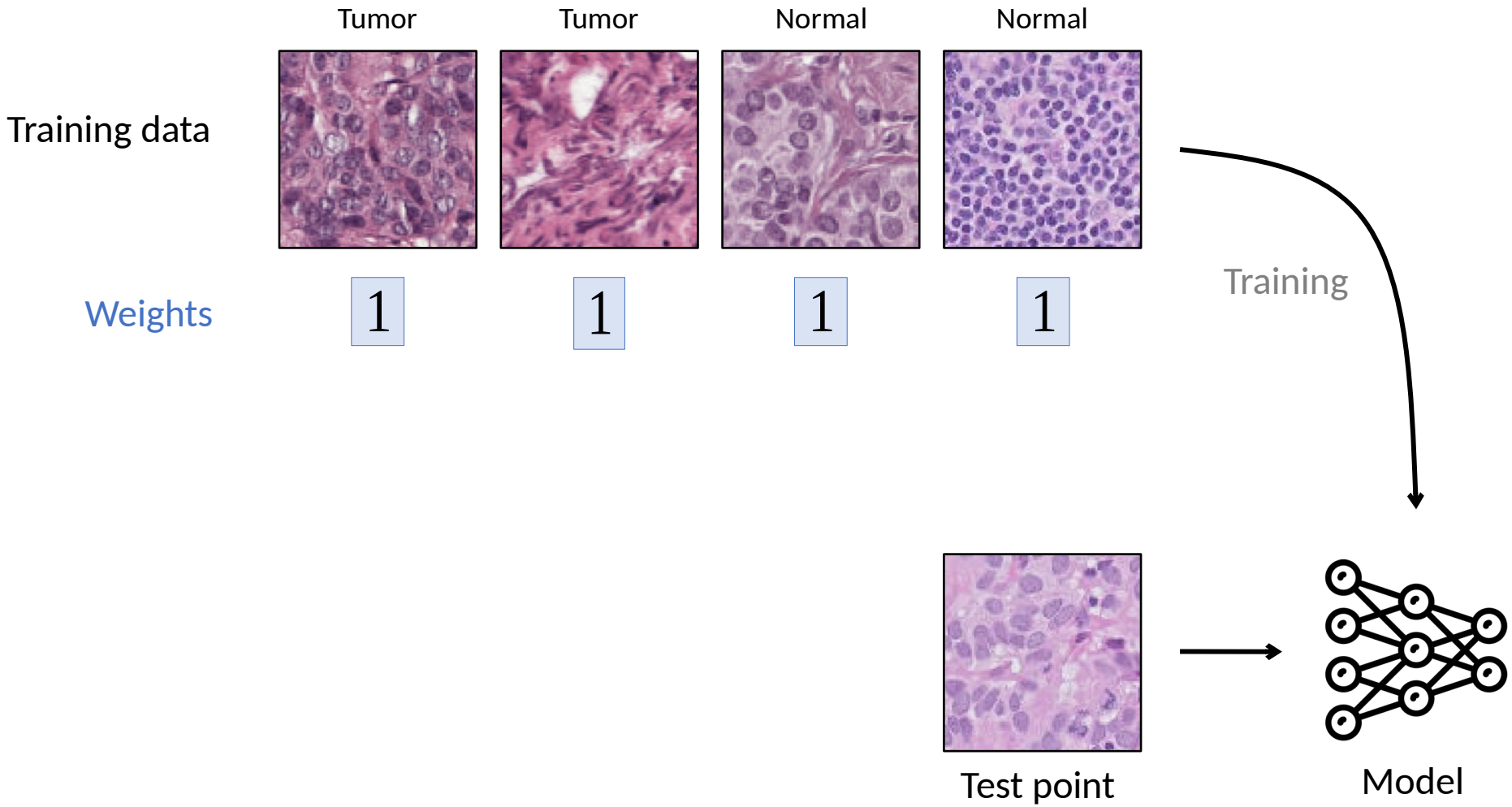
Problem

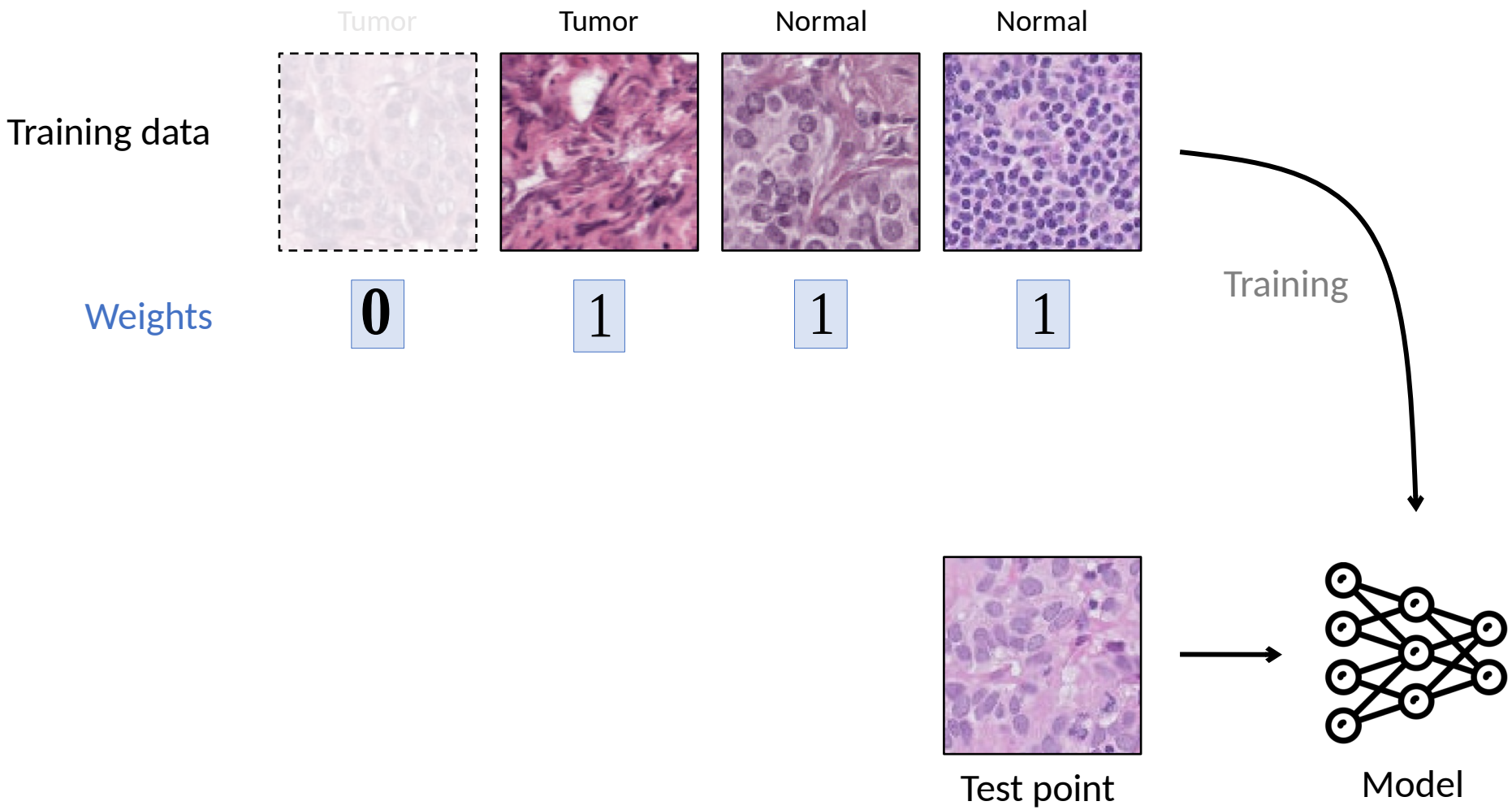
Repeatedly removing training points and retraining is too slow

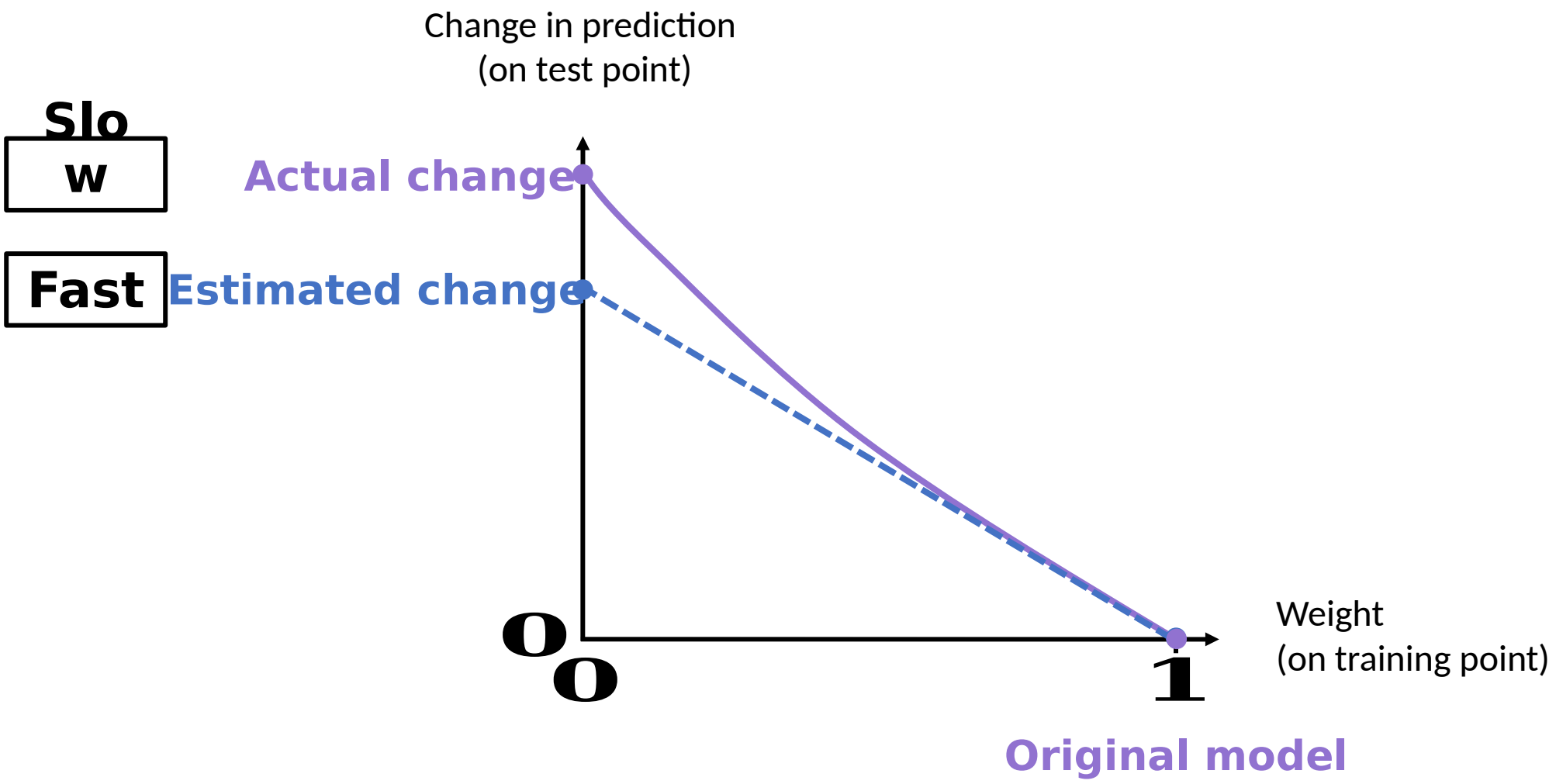
Solution

First-order Taylor approximation via influence functions




Tumor
Tumor
Normal
Normal
Training data
Weights
1
1
1
1
Training
Test point
Model


Tumor
Tumor
Normal
Normal
Training data
Weights
0
1
1
1
Training
Test point
Model


Change in prediction
(on test point)
Slo
w
Fast
Actual change
Estimated change
0
0
1
Weight
(on training point)
Original model

The influence function approximation

$$z_{test}, \hat{\theta}) \approx$$

Loss on (original)

The influence function approximation

Change in loss on after removing

$$\ell\left(z_{test}, \hat{\theta}_{-z_{train}}\right) - \ell\left(z_{test}, \hat{\theta}\right) \approx$$

Loss on (after removing)

Loss on (original)

The influence function approximation

Change in loss on after removing

$$\ell\left(z_{test}, \hat{\theta}_{-z_{train}}\right) - \ell\left(z_{test}, \hat{\theta}\right) \approx$$

The influence function approximation

Change in loss on after removing

$$\ell\left(z_{test}, \hat{\theta}_{-z_{train}}\right) - \ell\left(z_{test}, \hat{\theta}\right) \approx \nabla_\theta \ell\left(z_{test}, \hat{\theta}\right)^T H_{\hat{\theta}}^{-1} \nabla_\theta \ell\left(z_{train}, \hat{\theta}\right)$$

Gradient of loss on

Inverse of the Hessian

Gradient of loss on

The influence function approximation

Doesn't require retraining!

Change in loss on after removing

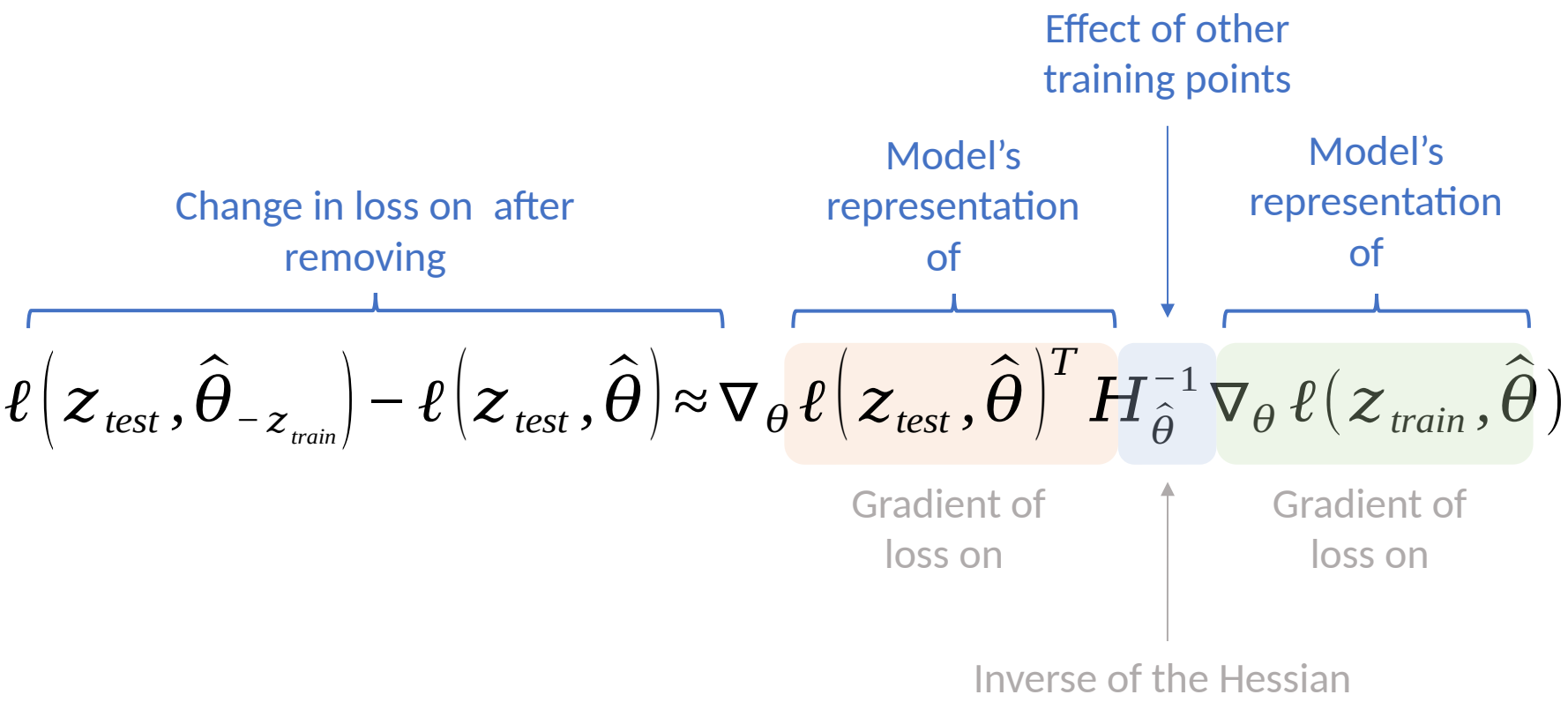
$$\ell\left(z_{test}, \hat{\theta}_{-z_{train}}\right) - \ell\left(z_{test}, \hat{\theta}\right) \approx \nabla_\theta \ell\left(z_{test}, \hat{\theta}\right)^T H_{\hat{\theta}}^{-1} \nabla_\theta \ell\left(z_{train}, \hat{\theta}\right)$$

Gradient of loss on

Inverse of the Hessian

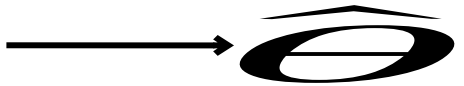
Gradient of loss on

The influence function approximation



A little bit more technical details ...

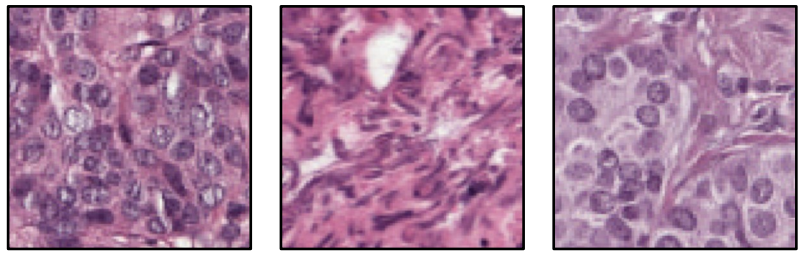
Training data



$\ell\left(z_{test}, \hat{\theta}\right)$

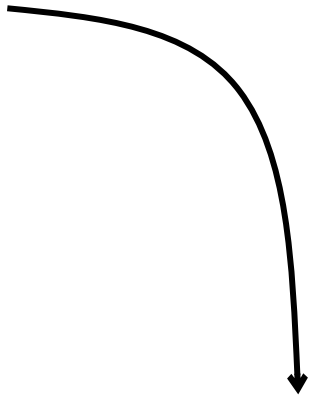
Training data

Tumor | Tumor | Normal



Weights

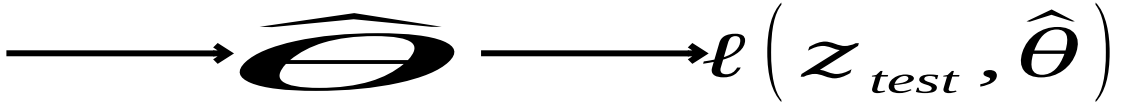
$+\epsilon$ | 1 | 1



$$\hat{\theta}_{\epsilon,z} \stackrel{\text{def}}{=} \arg\min_{\theta\in\Theta} \frac{1}{n}\sum_{i=1}^{n} L(z_i,\theta) + \epsilon L(z,\theta)$$

A little bit more technical details …

Training data

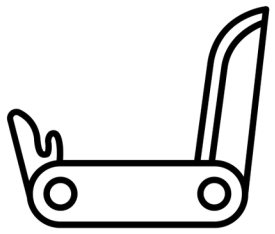


$$\left.\frac{d\hat{\theta}_{\epsilon,z}}{d\epsilon}\right|_{\epsilon=0} \qquad \left.\frac{dL(z_{\text{test}}, \hat{\theta}_{\epsilon,z})}{d\epsilon}\right|_{\epsilon=0}$$

$$= \nabla_\theta L(z_{\text{test}}, \hat{\theta})^\top \left.\frac{d\hat{\theta}_{\epsilon,z}}{d\epsilon}\right|_{\epsilon=0}$$

$$\hat{\theta}_{\epsilon,z} \stackrel{\text{def}}{=} \arg\min_{\theta\in\Theta} \frac{1}{n}\sum_{i=1}^{n} L(z_i,\theta) + \epsilon L(z,\theta)$$

From classical to modern settings



Jaeckel, 1972. The infinitesimal jackknife.

Hampel, 1974. The influence curve and its role in robust estimation.

Cook, 1977. Detection of influential observations in linear regression.

…

From classical to modern settings

Small datasets
Low-dimensional

Large datasets
High-dimensional

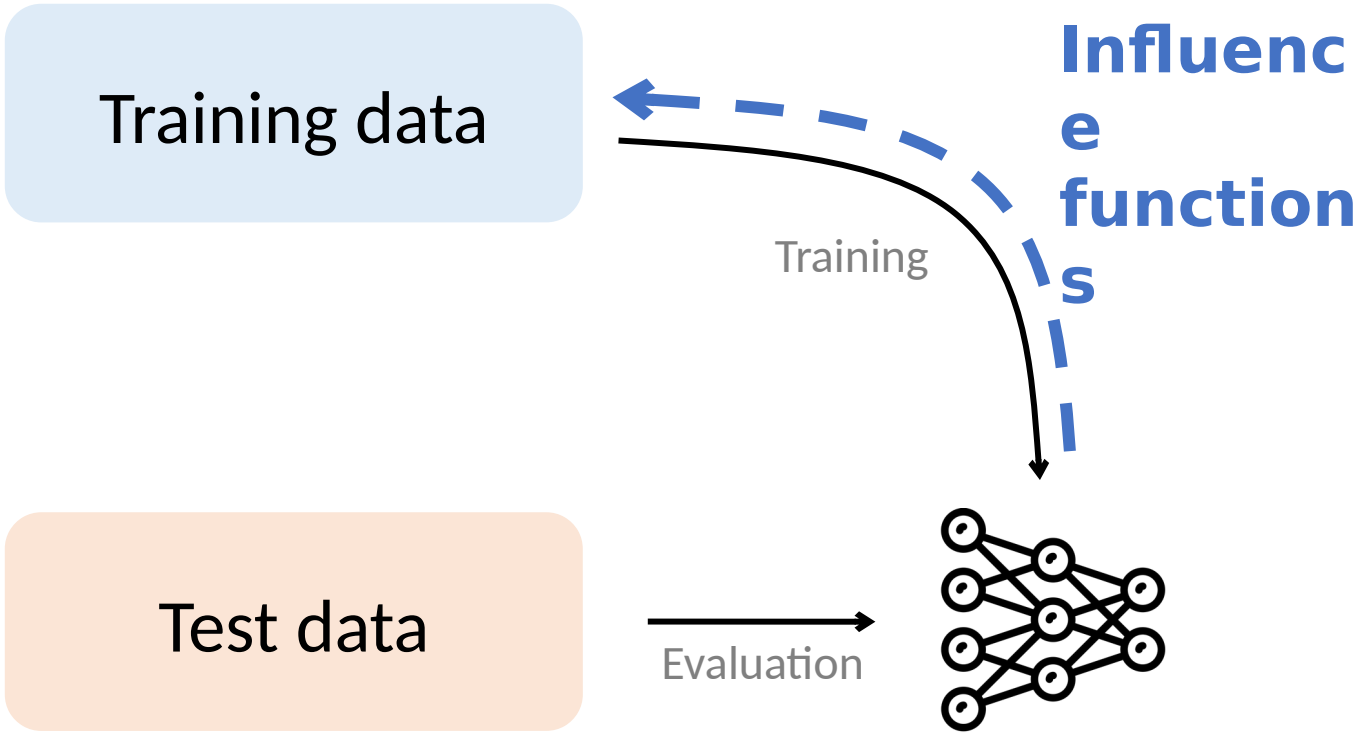
Difficult to compute

$$\nabla_\theta \ell\left(z_{test}, \hat{\theta}\right)^T \boldsymbol{H}_{\hat{\boldsymbol{\theta}}}^{-1} \nabla_\theta \ell\left(z_{train}, \hat{\theta}\right)$$

We use tools from
2nd-order optimization & stochastic estimation
[Pearlmutter, 1994; Martens, 2010, Agarwal et al., 2017]

Removing single points

Understanding Models via Their Training Data



Where can you apply influence functions?

Impact

Robustness

Training set biases
[Ren et al., 2018]

Inference reliability
[Broderick et al., 2021]

Cross-validation
[Stephenson et al., 2020]

Memorization
[Feldman, 2019]

Applications

Data distillation
[Wang et al., 2020]

Data valuation
[Jia et al., 2019]

Active learning
[Gudovskiy et al., 2020]

Data debugging
[Guo et al., 2021]

Fairness & security

Algorithmic bias
[Verma et al., 2021]

Data labor
[Arrieta-Ibarra, 2018]

Privacy
[Shokri et al., 2021]

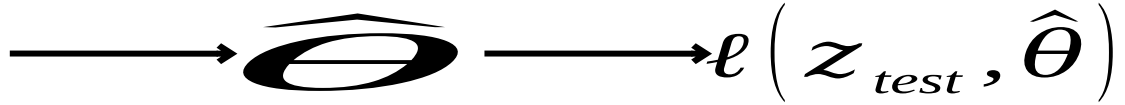
Data poisoning
[Chen et al., 2017]

Limitations

- What assumptions on the models are needed to calculate influence functions?
- Can you calculate influence functions for a neural network with the recipe described earlier?

$$\left.\frac{dL(z_{\text{test}}, \hat{\theta}_{\epsilon,z})}{d\epsilon}\right|_{\epsilon=0}$$

Training data



$$\hat{\theta}_{\epsilon,z} \stackrel{\text{def}}{=} \arg\min_{\theta\in\Theta} \frac{1}{n}\sum_{i=1}^{n} L(z_i,\theta) + \epsilon L(z,\theta)$$

Limitations

- What assumptions on the models are needed to calculate influence functions?
- Can you calculate influence functions for a neural network with the recipe described earlier?

If Influence Functions are the Answer, Then What is the Question?

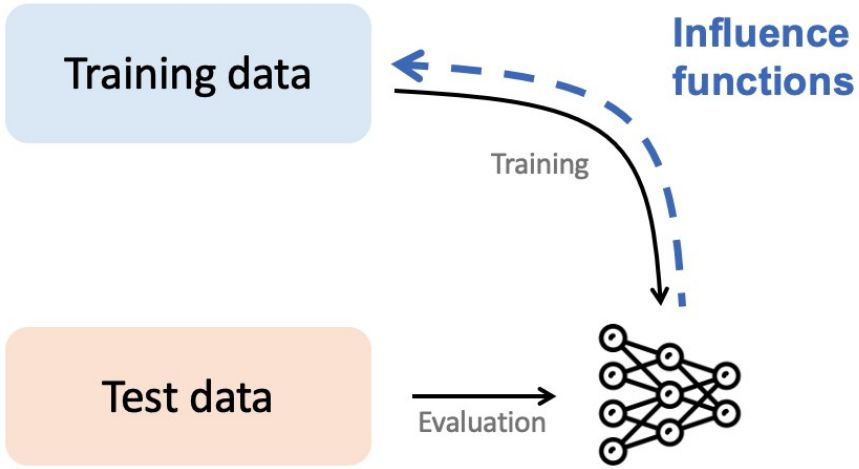
Juhan Bae*†, Nathan Ng†‡, Alston Lo†, Marzyeh Ghassemi‡, Roger Grosse†

Non-convexity Non-convergence

$$\hat{\theta}_{\epsilon,z} \overset{\text{def}}{=} \arg\min_{\theta\in\Theta} \frac{1}{n}\sum_{i=1}^{n} L(z_i,\theta) + \epsilon L(z,\theta)$$

Summary

- Link model behavior to training data
- Efficient to calculate
- Many applications



- Limitations when applying to complex models